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Paper 44
Entered: September 25, 2023

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ELI LILLY AND COMPANY,
Petitioner,

v.

TEVA PHARMACEUTICALS INTERNATIONAL GMBH,
Patent Owner.

IPR2022-00739
Patent 11,028,161 B2

Before JAMES A. WORTH, DAVID COTTA, and JAMIE T. WISZ
Administrative Patent Judges.

COTTA, *Administrative Patent Judge*.

DECISION
Final Written Decision
Determining All Challenged Claims Unpatentable
Denying-in-Part and Dismissing-in-Part Patent Owner's Motion to Exclude
35 U.S.C. § 318; 37 C.F.R. § 42.64

I. INTRODUCTION

On March 23, 2022, Eli Lilly and Company ("Petitioner")[1] filed a Petition to institute *inter partes* review of claims 1–30 of U.S. Patent No. 11,028,161 B2 (Ex. 1002, "the '161 patent"). Paper 1 ("Pet." or "Petition"). We instituted trial on September 27, 2022. Paper 9 ("Institution Decision" or "DI"). During trial, Teva Pharmaceuticals International GmbH ("Patent Owner")[2] filed a Patent Owner Response. Paper 20 ("PO Resp."). Later filings include Petitioner's Reply (Paper 25 ("Pet. Reply") and Patent Owner's Sur-reply (Paper 33 ("Sur-reply")). An oral hearing was held on July 19, 2023, and a transcript is entered in the record. Paper 43 ("Tr.").

We have jurisdiction under 35 U.S.C. § 6(b). After considering the parties' arguments and evidence, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

A. Related Matters

The parties indicate that they are involved in a district court litigation in *Teva Pharmaceuticals International GmbH et al. v. Eli Lilly and Company,* 1-21-cv-10954 (D. Mass.). Pet. 71; Paper 4, 1.

The parties also identify U.S. Patent Nos. 11,028,160 B2 ("the '160 patent") and 10,392,434 B2 ("the '434 patent") as relating to the '161 patent. Pet. 71–72; Paper 4, 1–2. The '434 patent, issued from U.S. Patent Application No. 15/712,444 ("the '444 application"), which claims priority to two provisional patent applications. Ex. 1045, code (21), (60). The '160

[1] Petitioner identifies itself as the real party-in-interest. Pet. 71.

[2] Patent Owner identifies itself and Teva Pharmaceuticals USA, Inc. as the real parties-in-interest. Paper 4, 1.

and ’161 patents both issued from applications filed as continuations of the ’444 application. IPR2022-00739, Ex. 1001, code (63); Ex. 1002, code (63). Thus, the ’160, ’161, and ’434 patents all share substantially the same disclosure and claim priority from the same provisional patent applications. The parties also identify U.S. Patent Application No. 17/308,580, which is currently pending before the Office, as claiming priority from the applications that issued as the ’160 and ’434 patents. Pet. 71; Paper 4, 2.

The ’160 patent is challenged in IPR2022-00738 and the ’434 patent is challenged in IPR2022-00796. Pet. 71; Paper 4, 1.

B. The ’161 Patent

According to the ’161 patent, “[m]igraine is a prevalent neurological condition characterized by attacks of headache and associated symptoms, such as nausea, vomiting, photophobia, and/or phonophobia.” Ex. 1002, 1:18–20. Preventative treatment may be appropriate “where frequency of attacks per month is two or higher, or where a patient’s quality of life is severely impaired.” *Id.* at 1:36–39. The ’161 patent discloses that “a number of drugs from different pharmacological categories (e.g. beta blockers, anticonvulsants) have been approved for migraine prevention or have class A evidence to support their use,” but explains that “response and tolerance to some of these medications varies, and compliance and adherence to these medications can be poor.” *Id.* at 1:40–46.

The ’161 patent teaches that calcitonin gene-related peptide (CGRP) “has been found to be involved in migraine processes, both centrally and peripherally.” *Id.* at 1:48–50. More specifically, the ’161 patent reports that “[j]ugular levels of CGRP are increased during migraine attacks, and intravenous (iv) CGRP administration induces migraine-like headache in most individuals with migraine.” *Id.* at 1:52–55. “CGRP is involved in the pathophysiology of migraine at all levels, peripherally (vasodilation, inflammation, and protein extravasation), at the trigeminal ganglion, and inside the brain." *Id.* at 1:57–60. According to the '161 patent, "[s]tudies have shown that inhibition of CGRP or antagonizing CGRP receptor has demonstrated efficacy in the treatment of EM [episodic migraine – i.e., migraine occurring less than 15 days per month]." *Id.* at 1:61–63. The '161 patent thus suggests that "[m]onoclonal antibodies that modulate the CGRP pathway . . . represent a class of promising therapeutic candidates for patients who failed prior preventative treatment for CM [chronic migraine] and EM." *Id.* at 2:1–4.

The '161 patent discloses "methods for preventing, treating, and/or reducing incidence of migraine in . . . a subject having refractory migraine by administering to the individual a therapeutically effective amount of an anti-CGRP antagonist antibody." *Id.* at 14:44–48.

C. Challenged Claims

The '161 patent includes thirty claims, all of which are challenged in the Petition. Claim 1, the only independent claim, is illustrative of the challenged claims and reads as follows:

> 1. A method of treating or preventing migraine in a subject having refractory migraine, the method comprising:
>
> selecting a subject who has been treated with two or more different preventative migraine treatments wherein at least one of the preventative migraine treatments is selected from topiramate, carbamazepine, divalproex sodium, sodium valproate, valproic acid, divalproex, flunarizine, candesartan, pizotifen, amitriptyline, venlafaxine, nortriptyline, duloxetine, atenolol, nadolol, metoprolol, propranolol, bisopropol, timolol, onabotulmumtoxin A, lisinopril, and oxeterone; and

> administering to the subject a therapeutically effective amount of a humanized monoclonal anti-calcitonin gene-related peptide (CGRP) antagonist antibody comprising the amino acid sequence of the heavy chain variable region set forth in SEQ ID NO: 63 and the amino acid sequence of the light chain variable region set forth in SEQ ID NO: 62.[3]

Ex. 1002, 173:1–19.

D. Asserted Grounds of Unpatentability

Petitioner asserts two grounds of unpatentability in this Petition (Pet. 19), which are provided in the table below:

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
1–30	103	Sun,[4] Dodick,[5] Sharma[6]
1–30	103	Dodick, Sharma

Petitioner relies on the declaration of Dr. Stefan Evers (Ex. 1101) and the declaration of Dr. Deborah Hay (Ex. 1239),[7] among other evidence.

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[3] There does not appear to be any dispute, and thus we find, that the heavy chain variable region set forth in SEQ ID NO: 63 and the amino acid sequence of the light chain variable region set forth in SEQ ID NO: 62 correspond to galcanezumab. Ex. 1101 ¶ 133.

[4] Sun et al., U.S. Patent Publication No. 2016/0311913 A1, published Oct. 27, 2016 (Ex. 1006, "Sun").

[5] Dodick et al., *Safety and efficacy of LY2951742, a monoclonal antibody to calcitonin gene-related peptide, for the prevention of migraine: a phase 2, randomized, double-blind, placebo-controlled study,* 13 Lancet Neurol., 885–92, published Aug. 11, 2014 (Ex. 1003, "Dodick").

[6] Sharma, International Patent Publication No. WO 2016/205037 A1, published Dec. 22, 2016 (Ex. 1007, "Sharma").

[7] The original version (Ex. 1233) of Dr. Hay's declaration was filed without an attestation under 18 U.S.C. § 1001. With our permission (Ex. 2093, 22:12–23:8), Petitioner filed a corrected version (Ex. 1239) of Dr. Hay's declaration addressing this omission. We understand all citations to Ex. 1233 in the parties' pleadings to refer to Ex. 1239. Herein we refer to Ex. 1239 rather than Ex. 1233.

Patent Owner responds with a declaration from Dr. Brian M. Grosberg, along with other evidence. Ex. 2037.

II. ANALYSIS

A. Legal Standards

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic, Inc. v. Avid Tech, Inc.,* 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")).

To show obviousness under 35 U.S.C. § 103 the differences between the subject matter sought to be patented and the prior art must be such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which that subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective indicia of nonobviousness when presented.[8] *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

In determining the level of skill in the art, we consider the problems encountered in the art, the art's solutions to those problems, the rapidity with

[8] Patent Owner does not direct us to objective indicia of nonobviousness in this case.

which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Patent Owner identifies the skill level of a person of ordinary skill in the art ("POSA") as follows:

> A POSA with respect to the '161 patent would have possessed a strong understanding of migraine and its treatments. A POSA would typically have a Ph.D. in a relevant field (e.g., neurobiology, neurology, or pharmacology) with several years of experience studying migraine or an M.D. with experience in a relevant field (e.g., neurobiology, neurology, or pharmacology) with several years of experience studying migraine or treating migraine patients. EX2037, ¶¶21-23. A POSA may be part of a multi-disciplinary team, drawing upon his or her own skills and taking advantage of certain specialized skills of others in the team, to solve a given problem. *Id.* For example, such a team may be comprised of an M.D. specializing in treating migraine patients and a neurobiologist specializing in studying migraine.

PO Resp. 6–7. Petitioner offers a similar definition of the POSA, with the principal differences being that: 1) Petitioner does not identify neurobiology and pharmacology as relevant fields for a POSA having an M.D., 2) Petitioner specifies that the POSA's experience also includes studying the role of CGRP in migraine, and 3) Petitioner does not propose that a POSA could also be part of a multi-disciplinary team. Pet. 11.

We find the additional exposition in Patent Owner's definition of the POSA helpful. Moreover, Patent Owner's definition appears to be consistent with the level of skill in the art reflected in the prior art of record and the disclosure of the '161 patent. We do not see a need to specify that the POSA would have experience studying the role of CGRP in migraine. Accordingly, for purposes of this Decision, we accept Patent Owner's proposed definition of the person of ordinary skill in the art. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) ("the prior art itself [may] reflect[] an appropriate level" as evidence of the ordinary level of skill in the art) (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985)). The differences between Petitioner's and Patent Owner's definitions of the POSA do not materially affect our decision. Put another way, we would reach the same decision under either definition.

C. Claim Construction

We interpret a claim "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b)." 37 C.F.R. § 42.100(b) (2020). Under this standard, we construe the claim "in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." *Id.*

The parties identify two claim limitations as needing construction: "a subject having refractory migraine," and "selecting a subject who has been treated with two or more different preventative migraine treatments." Pet. 12–16; PO Resp. 7–15. We address each claim limitation in turn.

1. <u>"A subject having refractory migraine"</u>

The phrase "a subject having refractory migraine" appears in the preamble to claim 1. Ex. 1002, 173:2–3. The parties agree that the preamble is limiting because it provides context and antecedent basis for terms in the body of the claim. Pet. 12 ("Claim 1's preamble . . . should be construed as limiting at least because it provides context and antecedent [basis] for the claimed 'therapeutically effective amount.'"); PO Resp. 9 ("There is no dispute that the preamble '[a] method of treating or preventing migraine in a subject having refractory migraine, the method comprising' is limiting.”). We agree that the preamble of claim 1 limits the claim for the reasons expressed by the parties. *Rapoport v. Dement*, 254 F.3d 1053, 1059 (Fed. Cir. 2001) (finding that the phrase “treatment of sleep apneas” in the preamble of an interference count was limiting because it provided antecedent basis for the phrase “to a patient in need of such treatment” in the body of the interference count); *see also Eli Lilly Co. v. Teva Pharmas Int’l GmbH*, 8 F.4th 1331, 1340–41 (Fed. Cir. 2021) (discussing construction of claims directed to “methods of using a composition for a specific purpose,” explaining, “while there is no bright-line rule for determining whether a preamble is limiting, we have generally construed statements of intended purpose in such method claims as limiting”).

We turn now to the meaning of the phrase “a subject having refractory migraine.” The parties agree that the ’161 patent expressly defines the phrase “a subject having refractory migraine.” Pet. 13; PO Resp. 9. In this regard, the ’161 patent states:

> As used herein, refractory migraine patients (or “subject having refractory migraine”) are considered refractory if they have a documented inadequate response (in a medical chart or by treating physician’s confirmation) to at least two preventive medications (from a different cluster, defined below). Refractory migraine patients can also be considered refractory if they have a documented inadequate response (in a medical chart or by treating physician’s confirmation) to two to four classes of prior preventive medications (from different cluster[s], as defined below), e.g., inadequate response to two classes of prior preventive medications, inadequate response to three classes of prior preventative medications, or an inadequate response to four classes of prior preventative medications.

Ex. 1002, 18:61–19:7. Thus, a “subject having refractory migraine” should be construed as: a migraine patient having a documented inadequate response (in a medical chart or by treating physician's confirmation) to at least two preventive medications from a different cluster. *Id.*

Our definition of "a subject having refractory migraine" includes the term "inadequate response." The Specification expressly defines this term as well. *Id.* at 19:8–26. The Specification states:

> Inadequate response is defined as: no clinically meaningful improvement per treating physician's judgement, after at least three months of therapy at a stable dose considered appropriate for migraine prevention according to accepted country guidelines, or when treatment has to be interrupted because of adverse events that made it intolerable by the patient or the drug is contraindicated or not suitable for the patient. The three-month period may not apply if the drug is intolerable or contraindicated or not suitable for the patient. For onabotulinumtoxin A, an inadequate response is defined as: no clinically meaningful improvement per treating physician's judgement, after at least six months of therapy at a stable dose considered appropriate for migraine prevention according to accepted country guidelines, or when treatment has to be interrupted because of adverse events that made it intolerable by the patient. Or, if onabotulinumtoxin A is a previous preventative medication, at least two sets of injections and three months should have passed since the last set of injections.

Id. Accordingly, the phrase "inadequate response" in our construction of "a subject having refractory migraine" should be understood to mean: 1) no clinically meaningful improvement per treating physician's judgment, after at least three months of therapy at a stable dose considered appropriate for migraine prevention according to accepted country guidelines (hereafter, "Group I patients"), 2) treatment has to be interrupted because of adverse events that made it intolerable by the patient (hereafter, "Group II patients"), or 3) the drug is contraindicated or not suitable for the patient (hereafter, "Group III patients").

Our definition of "a subject having refractory migraine" references medications from a "cluster." As used in our definition, the term "cluster" should be understood to refer to the "clusters" in the three sets of "clusters" set forth in the Specification. *Id.* at 19:27–35 (identifying a first set of clusters, including clusters A–E), 19:36–44 (identifying a second set of clusters, including clusters A–G), 19:52–62 (identifying a third set of clusters, including clusters a–e).

Patent Owner offers additional constructions, based largely on extrinsic evidence, for the terms "adverse event," "contraindicated," and "not suitable," as used in the Specification's definition of "inadequate response." PO Resp. 12–13 (citing Exs. 2003, 2004, 2005, and 2037). We do not find it necessary to construe these terms. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy'" (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))); *Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011) ("[C]laim terms need only be construed 'to the extent necessary to resolve the controversy'").[9] Although we do not find it necessary to construe the term

[9] In its briefing on claim construction, Patent Owner addresses whether three months of therapy is a standard in the art. PO Resp. 10. Similarly, Patent Owner addresses how the POSA would have understood the term "refractory migraine" *as used in the prior art*. *Id.* at 10–12. These arguments are directed not to what the claim terms mean, but whether they extend to obvious subject matter. Accordingly, we do not consider these arguments as part of our claim construction analysis. Instead, we consider these arguments in connection with applying the language of the claims to the cited art.

"adverse event," Patent Owner's arguments with respect to this term merit further discussion.

Patent Owner argues that we should also construe the term "adverse event" as that term is used in the Specification's definition of "inadequate response," proposing that we define it to mean: "a serious undesirable experience associated with the use of a treatment." PO Resp. 12.

Petitioner disagrees, asserting that Patent Owner's position that "the plain and ordinary meaning of 'adverse event' is a "***serious*** adverse event," and thus different from *Sun's* 'side effects' is facially implausible." Pet. Reply 21. Petitioner asserts that "[t]he Patent only requires interruption due to 'adverse events that ***made it intolerable***,' not life threat[en]ing." *Id.* at 21–22. Thus, according to Petitioner, the plain and ordinary meaning of "adverse events" is "***any*** untoward medical occurrence associated with the use of a drug in humans." *Id.* at 22.

Neither party cites intrinsic evidence bearing on the construction of "adverse event" beyond the portion of the Specification defining "inadequate response" as including when "treatment has to be interrupted because of adverse events that made it intolerable." Ex. 1002, 19:8–15; PO Resp. 12–13 (acknowledging that "[t]he '161 patent does not provide an express definition of an 'adverse event'" and citing only extrinsic evidence); Pet. Reply 21–23 (citing only extrinsic evidence); PO Sur-Reply 18–19 (citing only extrinsic evidence). Like the parties, we do not find any other intrinsic evidence bearing on the construction of "adverse event." Accordingly, we turn to the extrinsic evidence provided by the parties.

Patent Owner provides the testimony of Dr. Grosberg, who testifies that "a POSA would have accorded the term its ordinary and customary meaning in the prior art: ***a serious undesirable experience*** associated with the use of a treatment." Ex. 2037 ¶ 45 (emphasis added). As support, Dr. Grosberg cites an FDA publication entitled "What is a Serious Adverse Event?" *Id.* The publication Dr. Grosberg cites plainly states that an adverse event is "***any undesirable experience*** associated with the use of a medical product in a patient." Ex. 2004, 1 (emphasis added).[10] It then specifies that "[t]he event is serious and should be reported to the FDA when the patient outcome is: [one of eight categories of outcomes, including, for example, death, life threatening conditions, and hospitalization]." *Id.* By indicating that an "adverse event" is "any undesirable experience," and then specifying the conditions under which an adverse event is deemed "serious," the publication Dr. Grosberg cites makes clear that – contrary to Dr. Grosberg's and Patent Owner's proposed definition – not every adverse event is "serious."

The extrinsic evidence cited by Petitioner is consistent with the evidence cited by Dr. Grosberg that not every adverse event is a serious adverse event. For example, Petitioner cites a Guideline from the International Council for Harmonisation ("ICH") of Technical Requirements for Pharmaceuticals for Human Use. Pet. Reply 22 (citing Ex. 1210). Consistent with the evidence cited by Dr. Grosberg, it defines "adverse event" as: "[a]ny untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment." Ex. 1210, 8 (further explaining that an adverse event "can therefore be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom,

[10] All citations to exhibits in this opinion, other than those to patents and declarations, are to the page number added by the parties.

or disease temporally associated with the use of a medicinal (investigational) product"). Similarly, an FDA document titled "Guidance for Industry and Investigators" states: "Adverse event means any untoward medical occurrence associated with the use of a drug in humans, whether or not considered drug related." Ex. 1209, 6; *see also* Ex. 1208, 1 (FDA document entitled "IND Application Reporting: Safety Reports, using identical language to define "adverse event").

The preponderance of the evidence supports that the POSA would not understand the term "adverse event" to be limited to "serious" events. Ex. 2004, 1; Ex. 1208, 1; Ex. 1209, 6; Ex. 1210, 8. The only evidence to the contrary is the testimony of Dr. Grosberg. Ex. 2037 ¶ 45. Because Dr. Grosberg's testimony is inconsistent with the evidence he relies upon, we decline to give it substantial weight. We recognize that the Specification of the '161 patent includes language imposing some degree of severity on the "adverse event." Specifically, the adverse event must be significant enough to cause treatment to be "interrupted" because the event made the treatment become "intolerable by the patient." But, we find no support in the record for imposing an additional severity requirement by effectively limiting the term "adverse event" to "serious adverse events." Accordingly, we decline to construe the term "adverse event" to mean "a serious undesirable experience associated with the use of a treatment" as proposed by Patent Owner. PO Resp. 12. We do not find it necessary to further construe the term "adverse event" to resolve this dispute.

2. "Selecting a subject who has been treated with two or more different classes of preventative migraine treatments"

Claim 1 requires "selecting a subject who has been treated with two or more different preventative migraine treatments." Ex. 1002, 173:4–5. Hereafter, we refer to this claim language as the "selecting limitation." Petitioner contends that the selecting limitation does not require an inadequate response to treatment with the two different classes of medication, but rather requires only selecting "a subject to whom [two] medication[s] ha[ve] been administered." Pet. 15–16, 45–46 ("the claimed 'selecting' step only requires having 'been treated with'—but not necessarily ***failed***—multiple conventional treatments."). Petitioner's construction has particular relevance in the context of contraindicated patients. A patient that is contraindicated for a particular medication would not be treated with that medication. Ex. 2037 ¶ 46. However, such a patient could be administered medications that are not contraindicated. Thus, according to Petitioner, claim 1 encompasses selecting patients who are contraindicated for two migraine treatments – thus meeting our construction of the term "refractory" in the preamble – and are treated with, but do not have an inadequate response to, two classes of preventative migraine medications that are not contraindicated.

Patent Owner argues that the selecting limitation should be interpreted to require an inadequate response to two or more different classes of preventative migraine treatments. PO Resp. 14–15. According to Patent Owner, "while the specification defines patients having refractory migraine as including patients who are contraindicated for two or more treatments, the selecting limitation in the claims excludes those patients from the claimed method of treatment." *Id.* at 14–15. Patent Owner thus argues: "when the selecting limitation is read together with the 'subjects having refractory migraine' limitation, the claims require selecting a patient who *has been treated* with two or more classes of preventative migraine treatment, *and* who has had *inadequate response to those* two or more preventative migraine treatments". *Id.* at 15. Thus, under Patent Owner's proposed construction, the selecting limitation excludes contraindicated patients who are treated with, but do not have an inadequate response to, two classes of preventative migraine medications that are not contraindicated.

To help illustrate the parties' dispute, we provide the hypothetical example of a pregnant subject who regularly suffers migraines. Because of her pregnancy, the hypothetical subject is contraindicated for medication A & B. However, she can still be treated with medications C & D without experiencing an "inadequate response" as that term is used in the challenged patent. Under Petitioner's construction, the pregnant subject would qualify as "having refractory migraine" as claimed (based on being contraindicated to medications A & B), and, if selected for treatment with a mAb, would meet the "selecting" limitation (based on the treatment with medications C & D). Under Patent Owner's construction, the pregnant subject would not meet the limitations recited in the claims because the contraindicated patient could never be treated with the contraindicated medical treatments. In that light, the "selecting" limitation cannot be applied to contraindicated medications, thus excluding patients who only meet the "contraindicated" portion of the "inadequate response" patients from the scope of the claim.

We begin by considering the language of the claims. The selecting limitation is, on its face, straightforward. It simply requires selecting a patient that has been treated with two or more different classes of preventative migraine medications. Ex. 1002, 173:4–5. The parties do not disagree. PO Resp. 12 (agreeing that our construction of the selecting limitation in the Institution Decision to mean "selecting a subject that has been treated with two different classes of preventative migraine medications" "is correct"); DI 20–21.

On its face, the selecting limitation does not include language requiring failure of the "two or more different classes of preventative migraine treatments." Ex. 1002, 173:4–5. Indeed, the language of the selecting limitation requires only that the subject have been "treated" with two medications. *Id.* Patent Owner does not direct us to support in the Specification or in the prosecution history for construing the term "treated," as used in the selecting limitation, to require any particular result. PO Resp. 14–16. On this record, we do not discern a basis for requiring that the "treated" patients in the selecting limitation have achieved, or failed to achieve, a particular response.

Patent Owner contends that the requirement that "treated" patients have an inadequate response can be found in the claim language as a whole. Addressing our claim construction in the Institution Decision, Patent Owner argues:

> The Board notes that its construction does not require that the two preventative migraine medications achieve, or fail to achieve, any particular result when administered to the claimed subject. ID, 20-21. While the Board is correct to observe that the selection limitation *itself* does not impose this requirement , other claim language *does* require a particular result. Namely, the definition of refractory migraine *requires* that the claimed patient population have an inadequate response to two or more prior therapies. EX2037, ¶¶37-40. The claim language *as a whole* thus limits the patient subpopulation to patients who have been treated with (selection limitation), and

> who have had inadequate response to (refractory migraine definition), to two or more classes of preventative migraine treatments, given at a stable dose for at least three months. *Id.*

PO Resp. 15–16 (emphasis in original). We do not find this argument persuasive.

We recognize that claim 1 is directed to treatment of refractory migraine, which suggests that the patient treated is not responsive to other medications. But, as discussed *supra* § III.C.1, Patent Owner expressly defined "subject having refractory migraine" to include patients that are contraindicated for two preventative medications. Such patients would not take the medications they are contraindicated for, but could take other migraine medications. Ex. 2037 ¶ 46 (Dr. Grosberg testimony explaining that "a POSA would have concluded that a patient should not (and would not) be administered a medication if the medication is contraindicated"). Patent Owner's construction would exclude such patients from meeting the definition of the term "refractory." On this preliminary record, Patent Owner does not direct us to, and we do not discern, support in the language of the claims, the Specification, or the prosecution history for such a limiting construction.

The definition of "refractory" provided in the Specification, and incorporated in our construction of claim 1, can be harmonized with the "selecting limitation" without excluding contraindicated patients if one simply recognizes the possibility that contraindication patients can be administered migraine treatments that are not contraindicated. Considering the language of claim 1 as a whole, thus, does not convince us the claims should be read to exclude contraindicated patients.

Accordingly, we construe the selecting limitation to require selecting a subject that has been treated with two different classes of preventative migraine medications. We do not construe the selecting limitation to require that the two preventative migraine medications achieve, or fail to achieve, any particular result when administered to the claimed subject.

D. Priority

Petitioner asserts that claims 1–30 are not entitled to an effective filing date earlier than the actual September 22, 2017 filing date because "the claims lack adequate § 112(a) support in the September 2016 '180 Provisional [application]." Pet. 16.

In its Preliminary Response, Patent Owner argued that "[t]here is no basis for Patent Owner to address, or for the Board to decide, the priority date to which the claims are entitled" because Dodick, Sun, and Sharma – i.e., the prior art relied upon in Grounds 1 and 2 – "predate[] the earliest possible priority date for the challenged patent, making Petitioner's entire priority discussion irrelevant." Paper 6 ("Prelim. Resp."), 14–15.

In the Institution Decision, we accepted Patent Owner's assertion that it was not necessary for us to decide the priority date to which the claims are entitled. In doing so we noted that "in addition to Dodick, Sun, and Sharma, the Petition characterizes several references that help to support its arguments as 'prior art'" and explained that for purposes of the Institution Decision, we would "assume that Petitioner's characterization of these references as 'prior art' is correct." *See, e.g.*, DI 21 (citing, e.g., Exs. 1037–1040, 1057, and 1058).

Patent Owner's Response asserts that the Institution Decision correctly decided that it was not necessary to resolve Petitioner's priority challenge and that "[n]othing about the post-institution posture of this proceeding changes that assessment." PO Resp. 57 ("The Board correctly found that 'in the absence of a challenge to the prior art status of any of the prior art relied upon by Petitioner, … it is not necessary for us to decide the priority date to which the claims are entitled.'").

Given that Patent Owner continues not to challenge the prior art status of any prior art relied upon by Petitioner, we accept Patent Owner's assertion that it is not necessary for us to decide the priority date to which the challenged claims are entitled. For purposes of this decision, we assume that the references Petitioner characterizes as "prior art" are, indeed, prior art.

E. Overview of the Asserted Prior Art

1. Sun

Sun is a U.S. Patent Publication. Ex. 1006. Petitioner asserts that Sun is prior art under 35 U.S.C. § 102(a)(1) and (2). Pet. 9. Sun discloses "a method for preventing or reducing the occurrence of migraine headache in a patient in need thereof comprising administering to the patient an anti-CGRP receptor antibody or antigen-binding fragment thereof." Ex. 1006 ¶ 9. One anti-CGRP receptor antibody disclosed in Sun is AMG 334, which both parties identify as "erenumab." *Id.* ¶¶ 27–39, 254–295; Pet. 9; PO Resp. 5. Sun discloses that its anti-CGRP receptor antibody "can be administered in combination with an agent that interferes with the binding of the CGRP ligand to the CGRP receptor to prophylactically treat migraine headache," such as an "anti-CGRP antibody." Ex. 1006 ¶ 250 (citing WO 2011/156324, which, according to Dr. Evers, discloses galcanezumab (Ex. 1101 ¶ 51), as evidence that "[a]nti-CGRP antibodies are known in the art"). Sun discloses that in some embodiments, its anti-CGRP receptor antibody may be administered to a patient that has "failed or is intolerant to treatment with two different classes of migraine prophylactic agents" and in other embodiments, "the patient has failed or is intolerant to treatment with three different classes of migraine prophylactic agents." *Id.* ¶ 68.[11]

Sun discloses a Phase II study to evaluate the safety and efficacy of erenumab for the prevention of episodic migraine. *Id.* ¶¶ 266–272. The "results showed that AMG 334 when administered at a monthly dose of 70 mg, was efficacious in preventing episodic migraine, and AMG 334 had a safety/tolerability profile similar to placebo." *Id.* ¶ 272. "Subgroup analyses demonstrated that the efficacy of AMG 334 was similar regardless of . . . prior history of prophylactic medication use (FIG. 7B)." *Id.* ¶ 268. Sun discloses plans for a second Phase II study focusing on prevention of chronic migraine (*id.* ¶¶ 273–289), the results of which "are expected to show that in subjects with chronic migraine, AMG 334 dose-dependently reduces from baseline the monthly migraine days compared with placebo and the adverse event profile of AMG 334 is similar to placebo" (*id.* ¶ 281). Finally, Sun discloses plans for a Phase III study to evaluate the safety and efficacy of AMG 334 in migraine prevention in subjects with episodic migraine. *Id.* ¶¶ 290–295. "The phase 3 study results are expected to show that in subjects with episodic migraine, AMG 334 has a greater reduction from baseline in mean monthly migraine days compared to placebo." *Id.* ¶ 295.

[11] Sun defines failure to respond as a "lack of efficacy of the prophylactic agent in reducing the frequency, duration, and/or severity of migraine headache in the patient following a standard therapeutic regimen of the agent" including an "inability to tolerate the migraine prophylactic agent" or instances where the agent is contraindicated. Ex. 1006 ¶¶ 65–66.

2. Dodick

Dodick is a journal article published in the journal, Lancet Neurology. Ex. 1003. Petitioner asserts that Dodick is prior art under 35 U.S.C. § 102(a)(1). Pet. 7–8. Dodick discloses a "randomised, double-blind, placebo-controlled, phase 2 proof-of-concept study" to assess "the safety and efficacy of LY2951742 [galcanezumab][12] for the preventative treatment of migraine." Ex. 1003, 886. The study participants included "[m]en and women aged 18–65 years, with at least a 1-year history of migraine according to the International Classification of Headache Disorders (ICHD-II), and between four and 14 migraine headache days per month." *Id.* The study excluded patients who "fail[ed] to respond to more than two adequately dosed (ie, maximum tolerated dose by the patient for a sufficient duration) approved migraine prevention treatments." *Id.* Dodick reports that "LY2951742 was effective and generally well tolerated for the preventive treatment of migraine in patients with frequent attacks." *Id.* at 890. "Compared with placebo, LY2951742 significantly reduced the mean number of migraine headache days over the 3-month study period." *Id.*

3. Sharma

Sharma is an International Patent Publication. Ex. 1007. Petitioner asserts that Sharma is prior art under 35 U.S.C. § 102(a)(1) and (2). Pet. 9–10. Sharma discloses that "[a] phase [IIB], randomized, double-blind, placebo-controlled, dose-ranging study was conducted with 410 patients aged 18-65 years with 4 to 14 migraine headache days and at least 2 migraine attacks per month" to asses "whether at least one dose of

[12] There does not appear to be any dispute that LY2951742 is galcanezumab. Pet. 8; PO Resp. 5; Prelim. Resp. 26.

LY2951742 was superior to placebo in the prevention of migraine headache." Ex. 1007 at 23. "The results showed that all 4 dose arms were numerically superior to placebo on primary outcome measures at all-time points." *Id.* "One dose arm (120 mg) of LY2951742 met the primary objective (p = 0.004) with a significantly greater reduction compared to placebo in the number of migraine headache days in the last 28 day period of the week 12 treatment phase." *Id.*

F. Ground 1, Claim 1

Petitioner asserts that claims 1–30 would have been obvious over the combination of Sun, Dodick, and Sharma. We focus first on Petitioner's arguments with respect to claim 1. In brief summary, Petitioner contends that Sun discloses treating patients with "refractory migraine" using the anti-CGRP monoclonal antibody ("mAb"), erenumab. Pet. 20–28. To address the requirement of the challenged claims to treat refractory migraine with anti-CGRP mAb galcanezumab, Petitioner argues that it would have been obvious to use galcanezumab in the method disclosed in Sun, in view of Dodick's and Sharma's teachings that galcanezumab was effective for treating migraine. *Id.* at 28–33. Patent Owner opposes, arguing that Sun does not disclose treating refractory migraine patients, as that term is defined in the '161 patent, that a POSA would not have had reason to modify Sun, and that the cited art does not provide a reasonable expectation of success. *See generally* PO Resp.

We begin our analysis by considering whether Petitioner has established that Sun discloses treating refractory migraine patients with erenumab. We next consider whether Petitioner has articulated persuasive rationale to support that a POSA would have been motivated to use galcanezumab when treating refractory migraine patients. We then consider whether Petitioner established, by a preponderance of the evidence, that the POSA would have a reasonable expectation of success in treating refractory migraine patients with galcanezumab. Finally, we consider whether using galcanezuab in Sun's methods meets the limitations of claim 1 for all three patient groups identified in our claim construction.[13] We conclude that Petitioner has demonstrated by a preponderance of the evidence that claims 1–30 would have been obvious over the cited art.

1. Does Sun disclose treating refractory migraine patients?

As discussed in connection with claim construction, the '161 patent defines three categories of refractory patients: 1) patients who experience no clinically meaningful improvement per treating physician's judgment, after at least three months of therapy at a stable dose considered appropriate for migraine prevention according to accepted country guidelines ("Group I patients"), 2) patients for whom treatment had to be interrupted because of adverse events that made it intolerable by the patient ("Group II patients"), and 3) patients from whom a treatment is contraindicated or not suitable ("Group III patients"). Petitioner contends that Sun discloses treating each of these three categories of patients. Patent Owner contends that Sun does not disclose Group I and Group II patients, but does not dispute that Sun discloses Group III patients. We discuss each of these categories of patients

[13] Although Petitioner need only establish that it would have been obvious to treat one patient group, for completeness, this opinion addresses all three. *See* Tr. 7–8 (counsel for Petitioner agreeing that it is only necessary to show obviousness for one patient group), 43 (counsel for Patent Owner agreeing that it is not necessary to show that each patient group would have been obvious, but arguing that Group III patients are excluded from the scope of claim 1).

in turn. For the reasons discussed below, we find that Sun discloses all three categories of refractory migraine patient.

a) Group I patients (no clinically meaningful improvement)

Sun discloses that in some embodiments, its anti-CGRP receptor antibody may be administered to a patient that has "failed . . . treatment with two different classes of migraine prophylactic agents." Ex. 1006 ¶ 68; *see also id.* (disclosing that in other embodiments, "the patient has failed . . . treatment with three different classes of migraine prophylactic agents"). Sun defines the terms "failure to respond" and "treatment failure" to refer to "the lack of efficacy of the prophylactic agent in reducing the frequency, duration, and/or severity of migraine headache in the patient following a standard therapeutic regimen of the agent." *Id.* ¶ 65.

Petitioner contends that treating patients that have shown a "lack of efficacy" when administered prior therapies meets the definition of "no clinically meaningful improvement" that was recited in the Specification of the '161 patent (and adopted in our claim construction). Pet. 21–22. Petitioner addresses the requirement of our claim construction that "no clinically meaningful improvement" be determined "per treating physician's judgement, after at least three months of therapy at a stable dose considered appropriate for migraine prevention according to accepted country guidelines" by citing evidence that such criteria are "among common criteria for evaluating migraine prophylactics." *Id.* 22 n.8 (citing Ex. 1101 ¶¶ 24–25, 110 n.12; Ex. 1011, 3; Ex. 1019, 2; Ex. 1020, 4–5).

Patent Owner argues that the cited art does not disclose treating Group I patients because "Sun does not specify the duration of the dose at which the preventative medication is administered before it is considered a failure–an essential metric for diagnosing a patient as refractory under the ’161 patent definition.” PO Resp. 23. Patent Owner contends that “the art taught a wide variety of different durations” for assessing efficacy, and thus three months of therapy is “not a defined standard in the art.” PO Resp. 10–11 (citing prior art reflecting different treatment lengths as well as Dr. Grosberg’s testimony that three months is not a defined standard).

(i) The POSA’s understanding of Sun’s “standard therapeutic regimen”?

We begin our analysis of whether Sun discloses treating Group I patients by considering how the POSA would have understood the phrase “standard therapeutic regimen” in Sun’s definition of the terms “failure to respond” and “treatment failure” as it relates to duration of treatment. Ex. 1006 ¶ 65 (“As used herein, ‘failure to respond’ or ‘treatment failure’ refers to the lack of efficacy of the prophylactic agent in reducing the frequency, duration, and/or severity of migraine headache in the patient ***following a standard therapeutic regimen*** of the agent.”) (emphasis added). We find that the POSA would have understood the phrase “standard therapeutic regimen” to refer to a treatment regimen of as few as six weeks (i.e., 1.5 months).

The evidence supports Petitioner’s and Dr. Evers’s position that three months was a common minimum treatment length for assessing efficacy of a prophylactic migraine treatment.[14] Ex. 1101 ¶¶ 24, 25 (Dr. Evers’s

[14] Patent Owner argues that three months of therapy “is not a defined standard in the art.” PO Resp. 10. However, we do not understand Petitioner to contend that three months is a “defined standard.” Pet. 22–23 n.9 (asserting that three-month evaluation period was “among common criteria for evaluating migraine prophylactics”); Ex. 1101 ¶ 24 (Dr. Evers’s testimony that “[a] three month evaluation period was a commonly

testimony that three months was a "commonly recommended minimum duration" for evaluating efficacy); Ex. 1011, table 1 (table including three definitions of "refractory," one which states that "[a] ***3-month treatment period is required*** to assess efficacy but it may be useful to continue for a further 3–6 months if there was some improvement during the first 3 months,"); Ex. 1019, 2 (explaining that when beta blockers are administered for migraine prevention, "[a]n ***adequate trial of 3 to 12 months*** with continued assessment of efficacy and tolerability is recommended"); Ex. 1020, 4–5 ("A migraine prophylaxis is regarded as successful if the frequency of migraine attacks per month is decreased by at least 50% ***within 3 months***."); Ex. 1228, 15 (guidelines for clinical trials of migraine treatments recommending that "[t]reatment periods of at ***no less than 3 months*** in phase II RCTs [randomized controlled trials] and up to 6 months in phase III trials should be used"); Ex. 1224, 7 ("First, headache guidelines recommend assessing response to efficacy to prophylactic medications ***after 3 months of use***."); *see also* Ex. 1023, 5 ("Oral preventive medications must be titrated over weeks to effective doses, and then administered daily for ***approximately 3 months to establish efficacy***."); Ex. 1022, 2 (open-label study in which a patient was considered to have failed a prior treatment if that treatment was "***used in adequate doses for at least three months***") (all emphasis added).

However, the evidence also supports Patent Owner and Dr. Grosberg's position that the "requirement of 'at least three months of

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recommended minimum duration") ¶ 109, n.12 (Dr. Evers's testimony that "a period of three months at a stable dose was one of the commonly recommended treatment periods for evaluating the efficacy of the migraine prophylactic" and a "routine practice").

therapy at a stable dose' for Group I . . . patients was not standard in the art" and Patent Owner is correct that the prior art includes evidence showing patients "evaluated for response to migraine medications after *less than three months* of therapy." Ex. 2037 ¶ 43 (Dr. Grosberg's testimony); Ex. 1011, table 1 (table including three definitions of "refractory," one which states that an adequate trial is "typically ***at least 2 months*** at optimal or maximum-tolerated dose, unless terminated early due to adverse effects."); Ex. 1016, 5 ("continuing treatment for ***at least 2 to 3 months*** after the target dose is achieved is important to determine maximal efficacy"); Ex. 1013, 4, ("An adequate trial is defined as a period of time during which an appropriate dose of medicine is administered, typically ***at least 2 months*** at optimal or maximum-tolerated dose, unless terminated early due to adverse effects."); Ex. 1021, 6 ("Give each drug an adequate trial. ***It may take 2 to 3 months to achieve clinical benefit***.") (all emphasis added).

Of particular importance in interpreting Sun, Tepper, a reference that reports the results of one of the prospective clinical trials described in Sun, describes an "adequate trial" as "at ***least 6 weeks*** of treatment at generally accepted doses." Ex. 1037, 3 (emphasis added).

The record regarding how the POSA would have understood Sun's "standard therapeutic regimen" stands in near equipoise. The evidence discussed above strongly supports that three months is a very common, but not standard, recommended minimum treatment length. However, in the context of Sun's disclosure, we give greater weight to Tepper's disclosure of an evaluation period of "at least 6 weeks" than to the many disclosed three-month evaluation periods because Tepper's disclosure reports the results of a trial discussed in Sun. With the greater weight accorded to Tepper, we find that the POSA would not have understood Sun's "standard therapeutic regimen" to refer to a minimum of three months of treatment. Ex. 2037 ¶ 116 (Dr. Grosberg's testimony that Tepper "describes 'at least 6 weeks of treatment' as an adequate trial period for determining whether a patient had a therapeutic response" and thus that a POSA would have understood Example 4 of Sun to disclose a "prior preventative medication a failure after *at least six weeks* of therapy, rather than *at least three months* of therapy"). Rather, we find that the POSA would have understood the phrase "standard therapeutic regimen," as used in Sun, to refer to treatment durations of at least six weeks.

(ii) Does Sun render treating Group I patients obvious?

Having determined that the POSA would not have understood Sun's "standard therapeutic regimen" to refer to at least three months of failed treatment with a migraine prophylactic, and thus that Sun does not expressly disclose treating Group I patients, we next consider whether Sun suggests treating such patients.

The only difference identified by Patent Owner between Sun's patients who "failed . . . treatment with at least two different classes of migraine prophylactic agents" and the claimed Group I patients is how long the patients were administered treatments before those treatments were deemed failures; treatments of Sun's patients were administered for at least 6 weeks, while treatments of the claimed Group I patients must be administered for at least 3 months.[15] The evidence does not support that this

[15] In Patent Owner's Preliminary Response, Patent Owner argued that Sun's "failure to respond" is distinguishable from the '161 patent's "no clinically meaningful improvement" because the POSA would have understood Sun's "standard therapeutic regimen" to include "dose *adjustments*, such as a stepwise increase or decrease in dosage to find the optimum therapeutic

difference patentably distinguishes the claimed Group I patients from Sun's patients.

Petitioner argues that "it is immaterial whether three months at a stable dose was the ***only*** or 'a ***defined*** standard'" because Patent Owner's "few alternative evaluation periods—'at least two months,' 'two to six months,' 'two to three months,' 'at least six weeks' (POR, 29)—are at best 'a limited number of discrete permutations' that do not make 'at least three months' any less obvious." Pet. Reply 10. We agree.

As discussed above, the evidence establishes that it was common practice to administer a drug for a minimum of three months. *See supra* § II.F.1.a.i. In addition, the prior art of record uniformly encompasses, in the various definitions of "refractory" or "patients who failed treatment," patients whose treatments were given longer than six weeks to show results. Ex. 1019, 2 (3 to 12 months); Ex. 1020 (within 3 months); Ex. 1228, 15 (no less than 3 months in phase II RCTs and up to 6 months in phase III trials); Ex. 1224, 7 (after 3 months of use); Ex. 1023, 5 (approximately 3 months); Ex. 1022, 2 (at least 3 months); Ex. 1016, 5 (at least 2 to 3 months); Ex. 1013, 4 (at least 2 months); Ex. 1021, 6 (2 to 3 months); Ex. 1011, table 1 (providing two definitions, one requiring 3 months, the other requiring at least 2 months). Even Tepper, whose definition sets forth the lowest

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dose." Prelim. Resp. 23. This argument is not included in Patent Owner's Response and is, thus, waived. Paper 12, 9 ("Patent Owner is cautioned that any arguments not raised in the response may be deemed waived"). Even if we were to consider this argument, the evidence of record supports that although dose may be adjusted to determine an optimum therapeutic dose, a POSA would have understood to determine efficacy after treatment with a stable dose. *See* Ex. 1023, 5; Ex. 1101 ¶¶ 24–25, 109 n. 12; Pet. 22 n. 8; Ex. 2037 ¶ 89.

minimum treatment duration among the references identified by the parties, encompassed patients who failed treatments that were given more than six weeks to show results in defining "no therapeutic response" patients. Ex. 1037, 3 (excluding patients who had "no therapeutic response . . . with prophylaxis of more than three treatment categories . . . after an adequate trial (***at least*** 6 weeks of treatment at generally accepted doses)") (emphasis added).

These facts are analogous to those in *Prometheus Laboratories v. Roxane Laboratories*, 805 F.3d 1092 (Fed. Cir. 2015). In that case, the Federal Circuit affirmed the district court's finding that claims directed to a method for treating a species of patients were obvious over a prior art disclosure of a method for treating a genus encompassing that species. *Id.* More specifically, the claims at issue in *Prometheus* were directed to "treating a subset of those IBS [irritable bowel syndrome] patients — those who (1) are women (2) with IBS-D (3) who have experienced symptoms for at least six months and (4) who have had moderate pain" while the prior art disclosed treating the broader genus of patients with IBS. *Id.* at 1098. Of particular relevance here, the Federal Circuit found it obvious to treat patients who had experienced symptoms for at least six months on the basis that "it was common practice at the time . . . to determine whether a patient had suffered symptoms for longer than six months." *Id.* at 1099. Here, as in *Prometheus*, the claim at issue recites a limitation that was "common practice" with the effect of narrowing a genus patient population to a narrower subset of patients within the claimed genus. *See supra* § II.F.1.a.i.[16]

Patent Owner argues that the claimed Group I patients are more difficult to treat than Sun's patients and thus that the POSA would not expect success to "translate from Sun's disclosure to the claimed patients." PO Resp. 20, 23. This issue is better discussed in connection with our discussion of reasonable expectation of success. *See infra* § II.F.3.b.i. For purposes of our discussion here, it is sufficient to note: 1) that we do not agree with Patent Owner's position, and 2) that the evidence supports that the three-month observation period is not critical. Ex. 1196, 3 ("Although there is no evidence on the optimal length of prophylactic treatments, 3 months is usually considered sufficient to assess prophylactic efficacy,"); Ex. 1234, 198:18–19:9:13 (testimony of Dr. Grosberg: I agree [with a statement in Ex. 1196] that there was no evidence in the optimal length of prophylactic treatments at the time").

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[16] In *Prometheus*, the court also found that the claimed common practice of requiring patients to experience symptoms for longer than six months rather than just three months provided "a greater confidence in the diagnosis." 805 F.3d at 1099. Here, allowing patients to have more time to respond to treatment is similar in that it increases the confidence that they are truly refractory to that treatment. Ex. 2037 ¶ 59 (Dr. Grosberg testimony that "a patient who did not show improvement in a shorter evaluation period (e.g., 6 weeks) could still show improvement with the same medication after treatment for a longer period (e.g., 3 months) and perhaps was not even resistant to treatment in the first place."); *see also* Ex. 1013, 4 (recognizing that an evaluation period longer than 2 months "would be preferrable," but concluding that such an evaluation period would "prolong the time necessary to meet refractory criteria" and thus "prevent patients receiving the appropriate level of care.").

Considering all of the evidence and argument of record, including Sun's and Tepper's disclosures, the evidence establishing 3 months as a very common minimum evaluation period, the evidence supporting that all of the minimum evaluation periods identified in the art encompass patients who fail treatment after three months, and the evidence supporting that the 3 month evaluation period was not critical, we find that Sun's disclosure of treating patients that "failed . . . treatment with two different classes of migraine prophylactic agents" (Ex. 1006 ¶ 68) suggests using erenumab to treat patients who had failed at least three months of treatment with migraine prophylactics.

(iii) Conclusion with respect to Group I patients

In sum, we find that the only difference between Sun's patients who "failed . . . treatment with two different classes of migraine prophylactic agents" and the claimed Group I patients is how long the patients were administered treatments before those treatments were deemed failures. As to this difference, we find that Sun's disclosure of treating patients that "failed . . . treatment with at least two different classes of migraine prophylactic agents" render obvious an evaluation period of at least three months.

b) Group II patients (treatment interrupted by adverse events)

Petitioner contends that Sun's disclosure of treating patients who are intolerant to treatment with a particular migraine prophylactic agent and who discontinue treatment with a migraine prophylactic agent is a disclosure of treating Group II patients under our construction of "a subject having refractory migraine." Pet. 22–23; Pet. Reply 21–24.

Patent Owner disputes that Sun discloses Group II patients (i.e., patients whose "treatment has to be interrupted because of adverse events that made it intolerable by the patient"), contending that Sun instead discloses only "treating subjects with 'side effects.'" PO Resp. 30. Patent Owner argues that a POSA "would have known that 'side effects' are different from 'adverse events,'" and asserts that "Dr. Evers agrees." *Id.* (citing Ex. 2071, 148:24–150:16). According to Patent Owner "side effects" include "mild inconveniences such as morning grogginess, and even beneficial effects such as improved sleep" while adverse events "include serious events such as death, life-threatening circumstances, and permanent disability." *Id.* at 13. Thus, Patent Owner asserts, "[a] study that deems treatment a failure due to 'intolerable side effects' does not necessarily identify any adverse events." *Id.* at 30. We do not find this argument persuasive.

Sun discloses treating Group II patients. Sun discloses that in some embodiments, its anti-CGRP receptor antibody may be administered to a patient that has "failed . . . treatment with two different classes of migraine prophylactic agents." Ex. 1006 ¶ 68. Sun then discloses that "[f]ailure to respond to prior treatment with a migraine prophylactic agent can . . . include inability to tolerate the migraine prophylactic agent," such as when a patient "cannot tolerate the side effects associated with the agent." *Id.* ¶ 66. And Sun discloses that, in some embodiments, "a patient who has failed prior treatment with a migraine prophylactic agent is a patient who discontinues treatment with the migraine prophylactic agent due to associated side effects." *Id.* Sun's side effects are, in relevant part, the same as the claimed adverse events; both cause treatment to become intolerable and both result in treatment being interrupted. *Id.* ¶ 66; Ex. 1002, 19:8–15. We agree with Petitioner that Sun's disclosure meets the definition of Group II patients.

We are not persuaded by Patent Owner's argument (PO Resp. 30) that Sun does not disclose treating Group II patients because an "adverse event" is different from and more severe than Sun's "side effects." For the reasons discussed *supra* § II.C.1, we rejected Patent Owner's proposal that "adverse event" be construed as limited to "serious" events. The only requirement the claim imposes on the gravity of the adverse event is that it be severe enough to cause the patient to interrupt treatment. Sun's disclosure crosses that threshold.

We acknowledge Dr. Grosberg's testimony that "side effects were known in the art to be milder than adverse events." Ex. 2037 ¶ 45. We give this testimony some weight, but it does not persuade us that Sun's patients who cannot tolerate side effects do not qualify as Group II patients for two reasons. First, as discussed in connection with claim construction, Dr. Grosberg's testimony suggesting that "adverse events" must be "serious" is inconsistent with the evidence he cites to support his opinion. Ex. 2004, 1 (defining "adverse event" as "***any undesirable experience*** associated with the use of a medical product in a patient" and explaining when an "adverse event" is considered a "serious adverse event") (emphasis added). Second, his testimony that side effects include "mild inconveniences such as morning grogginess, and even beneficial effects such as improved sleep" (Ex. 2037 ¶ 45), even if true for some subset of "side effects" and some medical treatments, does not match the "side effects" at issue – those described in Sun's paragraph 66 (Ex. 1006 ¶ 66 (describing treating patients who "cannot tolerate the side effects" as well as patients who "discontinue[] treatment . . . due to the associated side effects"); *see also id.* ¶ 61 (defining the term "adverse side effect" as "any abnormality, defect, mutation, lesion, degeneration, harmful or undesirable reaction, symptom, or injury, which may be caused by taking the drug.”)).

Patent Owner’s argument that “Dr. Evers agrees” that “adverse events are more serious than side effects” (PO Resp. 30 (citing Ex. 2071, 148:23–150:16)) is unpersuasive because Dr. Evers’s testimony says nothing of the sort. Rather, consistent with the evidence cited by both parties supporting that “serious adverse events” are a subset of “adverse events” (Ex. 2004, 1; Ex. 1208, 1–2; Ex. 1209, 6, 9; Ex. 1210, 8, 13), Dr. Evers’s testimony addresses what adverse effects would be considered “serious adverse events” (Ex. 2071, 148:24–149:3 (explaining that “serious adverse event” is a “specific term” for when death occurs during a clinical trial), 150:3–151:11 (explaining that “[t]here is a list of serious adverse events in clinical trials, which comprises all events considered a serious adverse event,” and providing examples).

Patent Owner argues that Sun teaches that “intolerable side effects are those where the ‘impact’ is ‘greater than the therapeutic benefit of the migraine prophylactic agent.’” PO Resp. 24. According to Patent Owner “a POSA would not have considered such patients to have an inadequate response of the claimed method because Sun’s patients had a meaningful therapeutic benefit.” *Id.* This argument is not persuasive because it conflates Group I and Group II patients. While our construction of “inadequate response” requires that Group I patients exhibit “no clinically meaningful improvement,” our construction of Group II patients does not preclude patients from experiencing a benefit from the prior treatment; it requires only that patients have treatment “interrupted because of adverse events that made it intolerable by the patient.” *See supra* § II.C.1.

Dr. Grosberg points out that Sun uses both the term "adverse event" and the term "side effect." Ex. 2037 ¶ 98. Based on this fact, Dr. Grosberg asserts: "[b]ecause Sun chose to use 'side effects' to define 'failure to respond' (despite using 'adverse events' elsewhere), a POSA would have concluded that 'failure to respond' to a prior preventive medication in Sun's patients was because of *side effects* and not because of *adverse events*." *Id.* Although Patent Owner's Response cites this testimony (PO Resp. 30), the argument set forth in this testimony is not clearly presented in the briefing. While this violates our rules against incorporation by reference, we nonetheless consider it here. *See* 37 C.F.R. § 42.6(a)(3).

We do not find Dr. Goldberg's assertion persuasive because Sun's references to "adverse events" (Ex. 1006 ¶¶ 7, 61, 264, 268, 271, 281) do not diminish Sun's disclosure of treating Group II patients – i.e., patients who "cannot tolerate the side effects" of a migraine prophylactic treatment as well as patients who "discontinue[] treatment . . . due to the associated side effects" (*id.* ¶ 66). Moreover, in discussing that the present invention addresses the problem of poorly tolerated prior art therapies, Sun appears to use the terms "adverse events" and "adverse side effects" somewhat interchangeably. *Id.* ¶¶ 7, 8 (describing topimerate, a prior art anti-convulsant, as the most common migraine prophylactic, but teaching that there is "an urgent medical need for more effective and/or tolerable treatment options" after explaining that topimerate is "poorly tolerated" and that treatment can cause "adverse events"), 9 (describing the "present invention" as providing a migraine treatment "with no or minimal adverse side effects"), 14 (explaining that the disclosed treatment "does not substantially cause an adverse side effect associated with . . . antiepileptics"), 40 (explaining that current therapies "have a poor riskbenefit profile due to adverse side effects" and teaching that "[t]he present invention addresses this problem"); *see also id.* ¶ 61 (referencing "the Common Terminology Criteria for Adverse Events v4.0" in the paragraph that defines "adverse side effect"). Thus, Sun's separate use of the terms "adverse event" and "side effect" does not support that Sun's disclosure of treating patients with intolerable side effects (*id.* ¶ 66) is distinguishable from treating the claimed Group II patients.

In sum, we find that the evidence of record does not support the existence of a meaningful distinction between the Specification's definition of Group II patients and Sun's disclosure of treating patients who are unable to tolerate side effects. Accordingly, we find that Sun discloses treating Group II patients.

c) Group III patients (drug contraindicated or not suitable)

Sun discloses that in some embodiments, its anti-CGRP receptor antibody may be administered to a patient that has "failed . . . treatment with two different classes of migraine prophylactic agents." Ex. 1006 ¶ 68. Sun explains that the term "failure to respond" can include treating patients for whom the "side effects associated with the agent . . . may be incompatible with another medical condition which the patient has." *Id.* ¶ 66. Sun continues, "[b]y way of illustration, migraine prophylactic agents having a side effect of teratogenicity would be contraindicated in a pregnant patient." *Id.*

Petitioner contends that Sun's disclosure of treating contraindicated patients meets the requirements of treating Group III patients under our claim construction. Pet. 41.

In its Response, Patent Owner does not dispute that Sun discloses treating contraindicated patients, arguing only that the "selecting" clause excludes such patients from the scope of the claim. *See generally* PO Resp. (not challenging Petitioner's assertion that Sun discloses treating contraindicated patients).

In its Sur-reply, Patent Owner asserts that it "showed that Petitioner's cited art does not teach or suggest treating these [contraindicated] patients or provide any motivation to treat these patients with a reasonable expectation of success." PO Sur-reply 20 (citing Ex. 2037 ¶¶ 170–183, 200–203, 237–238, 249–250, 252–276). This argument is waived because it was not included in Patent Owner's Response. Paper 12, 9 ("Patent Owner is cautioned that any arguments not raised in the response may be deemed waived"). It also violates our rules prohibiting incorporation by reference. 37 C.F.R. § 42.6(a)(3); *Cisco Sys., Inc. v. C-Cation Techs., LLC*, IPR2014-00454, Paper 12 at 9 (PTAB Aug. 29, 2014) (informative). Accordingly, this argument is excluded from consideration.

Even if we were to consider this argument, we would not find it persuasive. The only challenge to Sun's disclosure of contraindicated patients in the long string cite provided by Patent Owner is in paragraph 200. That paragraph states that, for the reasons "explained above in Sections VI.b.iii [¶¶84–141] and VII [¶¶166–183], a POSA would have known that Sun does not teach or suggest the 'contraindicated' (i.e., Group III) patients of the '161 patent." Ex. 2037 ¶ 200. The gist of the argument in the testimony cross-referenced in paragraph 200 seems to be that Sun's use of the phrase "inability to tolerate" in connection with the term "contraindicated" means that the patients Sun describes as "contraindicated" must previously have been administered a medication. *Id.* ¶¶ 99–103. Thus, according to Dr. Grosberg, Sun's contraindicated patients would not qualify as "contraindicated" as that term is used in the '161 patent, because they had been treated with the supposedly contraindicated drug. *See id.* ¶ 102.

This argument is not persuasive for at least two reasons. First, we do not read Sun's use of the phrase "inability to tolerate the migraine prophylactic agent" to require that patients be treated with the intolerable drug. Ex. 1006 ¶ 66 ("Failure to respond to prior treatment with a migraine prophylactic agent can also include inability to tolerate the migraine prophylactic agent. . . . By way of illustration, migraine prophylactic agents having a side effect of teratogenicity would be contraindicated in a pregnant patient.")). A patient could be unable to tolerate side effects based on prior knowledge of how they will react to that treatment. For example, they may know that they are allergic to one of its ingredients or cannot tolerate one of its known side effects. Second, even if Sun's "inability to tolerate" requires prior treatment, the challenged claims do not place a restriction on when a patient becomes contraindicated. Put another way, the claims encompass as Group III patients, patients who are treated with a drug but subsequently become contraindicated for that drug – for example by becoming pregnant or being newly diagnosed with an unrelated medical condition during treatment.

In sum, even considering Patent Owner's waived arguments, we agree with Petitioner that Sun discloses treating Group III patients. *Id*. 1006 ¶ 66 ("[b]y way of illustration, migraine prophylactic agents having a side effect of teratogenicity would be contraindicated in a pregnant patient").

d) Conclusion regarding Sun's disclosure of the claimed patients

For the reasons discussed *supra* §§ II.F.1.a–c, we find that Petitioner has established, by a preponderance of the evidence, that Sun discloses or suggests treating each of Group I, Group II, and Group III patients.

2. <u>Has Petitioner articulated sufficient rationale to support that a POSA would have had reason to use galcanezumab to treat refractory migraine patients?</u>

Petitioner contends that the POSA would have been motivated to use galcanezumab, as disclosed in Dodick and Sharma, in Sun's methods for three reasons: 1) the POSA would have expected galcanezumab and erenumab to have similar efficacy, 2) Sun discloses combination therapy with galcanezumab and erenumab, and 3) Sun and Dodick disclose overlapping patient populations. Pet. 30–33. Patent Owner contests each of the three motivations proffered by Petitioner and argues, more generally, that the POSA, "would not have had any reason to jettison that successful drug and replace it with a *different drug* (galcanezumab) directed to a *different target* (CGRP), for a *different patient population* (selected refractory migraine patients of the claims), especially in view of the unpredictability surrounding the CGRP signaling pathway." PO Resp. 21–22. We address the evidence and the parties' arguments regarding motivation below. We find that Petitioner has established that the POSA would have been motivated to use galcanezumab in Sun's methods.

a) Expectation of similar efficacy

Petitioner contends that the a POSA would have understood Dodick and Sharma to teach "that galcanezumab was clinically at least similarly efficacious to *Sun's* erenumab," and, thus, it would have been obvious to substitute one anti-CGRP mAb (galcanezumab) for another (erenumab) to "effectively treat migraine by blocking the CGRP pathway." Pet. 31. According to Petitioner, this motivation was "reinforced by the expectation" that "blocking the CGRP pathway with clinically effective CGRP mAbs – whether by targeting the receptor (erenumab) or its ligand (galcanezumab) – would be similarly effective." *Id.* at 32. Petitioner asserts that "the similar efficacy of galcanezumab and erenumab" as well as the "potential for both to treat 'RM' [refractory migraine] patients" were "recognized in the art." *Id.*

Patent Owner disputes that similar efficacy provides motivation to use galcanezumab in Sun's methods on the ground that galcanezumab targets CGRP itself while erenumab targets its receptor. PO Resp. 20–22. In addition, Patent Owner disputes that similar efficacy provides motivation on the basis that other non-mAb treatments also targeted the CGRP pathway and, therefore, a POSA would not have sought to replace a drug that targets the CGRP pathway with another drug that targets the same pathway. We do not find these arguments persuasive. We address the evidence and arguments of record below.

(i) Evidence supporting similar efficacy of galcanezumab and erenumab

The expectation that galcanezumab would be efficacious in treating the claimed population is supported by studies showing that erenumab and galcanezumab have similar efficacy in the broader migraine population. *See* Pet. 31–33 (asserting that erenumab and galcanezumab have similar efficacy). For example, Sun discloses that erenumab is effective in treating migraine. Ex. 1006 ¶¶ 268–269 ("A statistically significant reduction in monthly mean migraine days was observed with AMG 334 70 mg (-3.40) vs placebo (-2.28). . . . Statistically significant reductions in monthly headache days (70 mg: -3.54 vs placebo: -2.39: P=0.022) and monthly acute migraine-specific medication use days (70 mg: -1.64 vs placebo: -0.69; P=0.004: FIG. 6) were also observed."). And Dodick and Sharma teach that galcanazumab is effective in treating migraine. Ex. 1003, 6 (reporting that galcanezumab was "effective and generally well tolerated for the preventive treatment of migraine in patients with frequent attacks" and "significantly reduced the mean number of headache days over the 3-month study period"); Ex. 1007, 23 (showing that all four dose arms of galcanezumab tested were "numerically superior to placebo on primary outcome measures at all-time points" and that one dosage (120 mg) showed a "significantly greater reduction compared to placebo in the number of migraine headache days."). Dr. Evers testifies, that the efficacy results reported in the studies are "comparable" as are the side effect profiles. Ex. 1101 ¶¶ 157–158 (comparing responder rate and reduction in monthly migraine headache days as compared to placebo both erenumab and galcanezumab). This testimony is consistent with the references and credible.

Dr. Evers also testifies that "[t]he comparable efficacy of galcanezumab and erenumab in the treatment of migraine was recognized throughout the art." *Id.* ¶ 159; *see also* Pet. 32 (asserting that "[t]he similar efficacy of galcanezumab and erenumab . . . was recognized in the art, which addressed use of the clinically proven anti-CGRP mAbs collectively"). We credit Dr. Evers's testimony on this point because it is consistent with a wealth of prior art references addressing anti-CGRP mAbs as a class and supporting that, among the general migraineur population, all of the members of this class (which includes erenumab, galcanezumab, fremanezumab, and eptinezumab) have similar efficacy and safety. *See* Ex. 1230, 2 (teaching that the four mAbs that have been subject to clinical trials "have been almost similarly effective, tolerable and safe in phase 2 studies and are being studied in phase 3 trials for episodic and chronic migraine prevention."); Ex. 1074, 9 ("based on the results of 5 Phase II trials, this review and meta-analysis revealed a significant effect of CGRP-mAbs for migraine prevention with few adverse reactions"); Ex. 1231, 4 ("The present meta-analysis [of four phase II clinical trials] demonstrates that CGRP mAbs lead to improvement in decrease of monthly migraine days from baseline to week 1–4 or week 9–12 after CGRP mAbs administrated and well tolerated, as compared with placebo."); Ex. 1232, 7 ("All four anti-CGRP mAbs investigated to date have led to significant reductions from baseline in either episodic and/or chronic migraine days per month compared with placebo."); Ex. 1037, 8 ("Results from five other phase 2 studies have been published on the efficacy and safety of monoclonal antibodies targeting the CGRP pathway for migraine prevention, all of which showed efficacy, and none of which raised safety concerns."); Ex. 1049, 10 (discussing clinical trials of anti-CGRP mAbs, including erenumab and galcanezumab, stating "[t]he numerous studies so far conducted with the available anti-CGRP monoclonal antibodies have shown satisfactory safety and efficacy outcomes in migraine prevention," and concluding that "the overall profile of anti-CGRP mAbs so far shown can be regarded as highly favorable" and that "in the forthcoming years, anti-CGRP mAbs will probably equal, in preventative treatment, the revolution introduced by triptans in acute treatment of migraine"); Ex. 1028, 2, 8 (article reviewing the "current state of development for mAbs targeting the CGRP pathway," concluding that "mAbs targeting the CGRP pathway are a promising new drug class that may provide a valuable new option for clinicians aiming to relieve the burden of individuals with episodic or chronic migraine."); Ex. 1025, 3 ("Overall, CGRP-mAbs look like promising options for migraine and chronic migraine prevention with impressive responder rates, improved safety and tolerability, absence of liver toxicity and long half-lives leading to infrequent dosing."); Ex. 1027, 5 ("Preliminary data showed positive results for all four mAbs."); Ex. 1042, 1 ("Monoclonal antibodies against CGRP or the CGRP receptor have a longer duration of action [that CGRP receptor antagonists] and have been investigated for migraine prevention. Four are in development and three have completed phase II and one phase III trials; every reported study has been positive. Furthermore, no safety issues have arisen to date, including hepatic or cardiovascular effects, and initial tolerability appears to be excellent."); Ex. 1036, 1–2, 6 (explaining that "[s]everal recent phase 2 studies with CGRP monoclonal antibodies [including galcanezumab and two other mAbs]. . . showed clinical benefit for migraine prevention, without hepatotoxicity concerns" and reporting that phase 2 studies of the mAb, erenumab, "showed a significant reduction in monthly migraine days from baseline versus placebo"); Ex. 1029, 6, 7 (opining that "[f]ully humanized mAbs targeting CGRP or its receptor directly appear more promising [than CGRP receptor antagonists] for the prophylactic treatment of frequent episodic and chronic migraine" and that "CGRP-targeted mAbs could provide a possibility for successful therapy in this field"); Ex. 1025, 3 ("Overall, CGRP-mAbs look like promising options for migraine and chronic migraine prevention with impressive responder rates, improved safety and tolerability, absence of liver toxicity and long half-lives leading to infrequent dosing.").

In addition to supporting that mAbs have comparable efficacy among the general migraine population, the prior art also supports the recognition that both erenumab and galcanezumab have the potential to treat refractory migraine. For example, a *post hoc* analysis of the results of a phase II study of erenumab, concluded that "erenumab 70 mg and 140 mg reduced the number of monthly migraine days with a safety profile similar to placebo." Ex. 1037, 1. The study included "453 (68%) patients who had failed at least one previous preventive drug class because of lack of efficacy or poor tolerability and 327 (49%) who had failed at least two previous preventive drug classes," which the author found to suggest that "there could b[e] efficacy in a treatment-resistant population." *Id.* at 8. Similarly, a *post hoc* analysis of a phase II study of galcanezumab concluded that the subgroup of patients with a "history of failure to preventative treatments" exhibited a "statistically significantly greater treatment effect (galcanezumab vs placebo difference) at Month 3," which the author found to suggest that this subgroup may be a "predictor[] of clinical response with greater treatment effects in episodic migraineurs." Ex. 1072, 74. Consistently, at least one reference suggests that the entire class of mAbs may have efficacy in treating refractory migraine. Ex. 1023, 7 ("No doubt, monoclonal antibodies (mAbs) against calcitonin gene-related peptide (CGRP) for preventative treatment of episodic and chronic migraine deserve to be called a breakthrough – not because they cure headache, but rather because they are effective for relatively refractory headaches"); *see also* Ex. 1053, 13 (suggesting that "mAbs targeting CGRP or its receptor hold the most promise for preventive treatment" of patients in the "medically refractory subgroup[]").[17]

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[17] A press release from Patent Owner provides support for the proposition that the entire class of mAbs may be effective in refractory patients by identifying another anti-CGRP mAb that may be effective in "highly

In sum, the evidence of record, including clinical trials of both erenumab and galcanezumab, Dr. Evers's credible testimony on the comparability of those clinical trials, and the wealth of prior art references addressing anti-CGRP mAbs as a class and supporting that all members of that class have similar efficacy, provide strong evidence that the POSA would have expected erenumab and galcanezumab to have similar efficacy and tolerability.

(ii) Evidence regarding CGRP pathway uncertainty and regarding differences between galcanezumab and erenumab in mechanism of action

Patent Owner argues "[g]alcanezumab is an entirely different drug than erenumab" comprised of a "completely different antibody, directed against a different target." PO Resp. 20. According to Patent Owner, "[a] POSA would not assume its effectiveness in a particular patient population merely because erenumab demonstrated efficacy in that population." *Id.* (citing Ex. 2037 ¶ 199). Indeed, Patent Owner asserts, "the POSA would have considered the CGRP pathway to be unpredictable in part due to CGRP's promiscuity with cellular receptors other than the CGRP receptor." *Id.* at 20–21. Patent Owner explains:

> As Walker explained in 2013, "[t]he physiological and pathophysiological actions of CGRP could be mediated [through] multiple receptor subtypes, including the CGRP receptor, the AM_2 receptor and the AMY_1 receptor." EX2046, 3; *see also*, EX2047, Abstract (identifying the "presence of two

refractory patients," fremanezumab. Ex. 1041, 2 ("The very fast onset of preventative response, seen after a single dose of therapy, along with the impressive decrease in migraine days, amongst such highly refractory patients, may bring us a step closer to provide widespread relief to people who suffer from chronic and episodic migraine.").

> CGRP-responsive receptors … AMY_1 … and the CGRP receptor.”); EX2037, ¶¶217-218. CGRP’s receptor promiscuity, coupled with “the complex nature of CGRP action,” made “understanding [CGRP’s] underlying biology and [signaling] mechanisms a distinct challenge.” EX2046, 10. This unpredictability in the art is reflected in Petitioner’s own references. For example, Dodick expressly discloses that “the site and mechanism of action of CGRP monoclonal antibodies is *unclear*.” EX1003, 6. Teva Press Release discloses that “long-term total disruption to the normal physiological functions of the CGRP system … are *unknown*.” EX1041, 3. And Tepper discloses that “the specific pathogenic mechanism of CGRP in migraine is *unknown*, including *uncertainty* with respect to its site of action.” EX1037, 8; EX2037, ¶¶217-218.

Id. (alterations in original).

We recognize that the evidence cited by Patent Owner, including two papers authored by Petitioner’s witness, Dr. Hay, support the possibility that CGRP activity may be mediated through multiple receptor subtypes. Ex. 2046, 3; Ex. 2047, 2; *see also* Ex. 1239 ¶¶ 22–23 (testimony of author, Dr. Hay, acknowledging that articles she authored propose that there may be multiple CGRP receptors). We also recognize the evidence that there are aspects of CGRP’s function – such as its “specific pathogenic mechanism . . . in migraine, including . . . its site of action” that were not known. Ex. 1037, 8; *see also* Ex. 1041, 3. However, we agree with Dr. Hay that any uncertainty created by the potential that there are multiple CGRP receptors and/or that CGRP has an uncertain mechanism of action would have been outweighed by the wealth of evidence (discussed *supra* § II.F.2.a.i) that erenumab and galcanezumab, indeed all these CGRP mAbs, have comparable safety and efficacy. Ex. 1239 ¶ 25 (“[A]ny inferences drawn from my papers in 2015 regarding potential differences in clinical properties among anti-CGRP pathway mAbs would have been outweighed by

September 2017 by the clinical trial data showing that erenumab, galcanezumab, and fremanezumab were comparable in terms of efficacy, tolerability, and safety."); *see also* Ex. 1034, 2 ("The expectation is that . . . ***antibodies against both the ligand and receptor would prevent CGRP-induced activation*** of sensitized central trigeminal pathways, therefore decreasing headache frequency over time."); Ex. 1035, 10 ("The introduction of ***mAbs targeting the CGRP neuroactive peptide and/or its main receptor*** appears to lay the foundation for a new class of prophylactic drugs that could finally overcome, even only partially, the efficacy, safety, tolerability, and adherence issues that often affect chronic migraineurs."); Ex. 1053, 13 (suggesting that "mAbs targeting ***CGRP or its receptor*** hold the most promise for preventative treatment" in specific subgroups of migraine patients, including the "medically refractory subgroup[]") (all emphasis added).

Patent Owner argues that "at the time of the invention, there were known advantages of targeting the CGRP receptor with erenumab." PO Resp. 22. As support, Patent Owner cites Dr. Grosberg's testimony that "it was known that 'high selectivity' for blocking the CGRP receptor itself was advantageous because . . . if . . . other receptors are blocked by a less selective agent, undesired side effects could arise." Ex. 2037 ¶ 218 (citing Ex. 2049, 2); PO Resp. 22. Patent Owner and Dr. Grosberg also cite several references teaching that the "ability to block the CGRP receptor (as opposed to CGRP itself) might be *advantageous* since binding to the CGRP receptor might prevent receptor activation, independent of CGRP release." *Id.* (citing Ex. 2048, 3; Ex. 2075, 8; and Ex. 2076, 4). According to Patent Owner and Dr. Grosberg, a POSA would not have "expected to see these same advantages" using galcanezumab and thus, would have been "dissuaded . . .

from switching drugs." PO Resp. 22; Ex. 2037 ¶ 218; *see also* PO Resp. 39 (arguing that "the art taught that targeting the CGRP receptor (such as with erenumab) was *advantageous* over targeting the CGRP ligand (such as with galcanezumab)").

We are not persuaded that a POSA would have perceived erenumab to be advantageous as compared to galcanezumab on the basis that galcanezumab is a "less selective agent" than erenumab. Patent Owner and Dr. Grosberg cite Shi 2016 (Ex. 2049) as teaching the "high selectivity" of erenumab for blocking the CGRP complex as compared to "less selective agents," and use this teaching to imply that galcanezumab is less selective than erenumab. PO Resp. 22; Ex. 2037 ¶ 218. But, as Dr. Hay explains, "[t]he 'less selective agents' that *Shi 2016* refers to are the small-molecule CGRP receptor antagonists, Gepants, not galcanezumab." Ex. 1239 ¶ 29. We find Dr. Hay's testimony on this point persuasive, as it is consistent with Shi 2016. *See* Ex. 2049, 2 (discussing concerns regarding small molecule CGRP antagonists that have prevented their approval and opining that "[a] monoclonal antibody may be a preferred modality to target the CGRP receptor"). Moreover, Shi 2016 concludes with a discussion of the advantages of the superior selectivity of mAbs as a class. *Id.* at 8 (discussing advantages of mAbs over small molecule CGRP-receptor antagonists, noting "[i]n addition to high potency and superior selectivity, which may provide the benefit of fewer off-target side effects, monoclonal antibodies provide advantages that make them better suited for preventive treatment compared with small molecules"); Ex. 1239 ¶ 32 (interpreting Shi as teaching that "antibodies as a class have limited off-target effects as compared to small molecules because they bind to fewer biological targets and are therefore considered more specific."). Accordingly, the evidence of record does not support that the POSA would have perceived erenumab to have a selectivity advantage over galcanezumab.

As to Patent Owner's argument that the POSA would have perceived erenumab to have the advantage that it "might prevent receptor activation, independent of CGRP release," we recognize that the references cited as support posit preventing receptor activation independent of CGRP release only as a theoretical advantage. *See* Ex. 2048, 3 ("***[i]n theory***, the ability to block the CGRP receptor (as opposed to CGRP itself) ***might be*** advantageous since binding to the CGRP receptor ***might*** prevent receptor activation, independent of CGRP release."); Ex. 2075, 8 (similar); and Ex. 2076, 4 (similar); *see also* Ex. 1239 ¶¶ 33–37 (Dr. Hay testimony discussing absence of data supporting this advantage). Nonetheless we give this theoretical advantage some weight in considering whether a POSA would have had reason to substitute galcanezumab for erenumab.[18] That said, we find that, like Patent Owner's evidence of uncertainty regarding the CGRP pathway, this theoretical advantage is outweighed by evidence from

[18] In determining whether Petitioner has established motivation to use galcanezumab in place of erenumab, we consider this theoretical advantage together with all of the evidence of record, including the evidence, discussed above, that the comparable efficacy of galcanezumab and erenumab was "recognized throughout the art." We are not persuaded by Patent Owner's argument that Petitioner cannot "discount Patent Owner's evidence that blocking the CGRP receptor would have been thought superior to binding the ligand, by arguing that the evidence was eventually disproven through later clinical trials" because "[o]bviousness turns on what a POSA would have expected *ex ante*—not what is later proven empirically." PO Sur-reply 9. In this regard, we note that the relevant date is not the date of Patent Owner's evidence supporting that blocking the CGRP receptor would have been thought superior, but rather the priority date of the challenged claims, which we discussed *supra* § II.D.

actual clinical trials (discussed *supra* § II.F.2.a.i) as well as suggestions in the art that targeting both the ligand or its receptor would be effective. *See* Ex. 1034, 2; Ex. 1035, 10; Ex. 1053, 13.[19]

In sum, we are not persuaded by Patent Owner's evidence and argument that unpredictability in the CGRP pathway and differences between targeting the CGRP receptor as compared to its ligand materially detracts from the expectation that galcanezumab and erenumab would have similar efficacy or otherwise negatively impacts a POSA's motivation to substitute one for the other.

(iii) Evidence regarding prior art non-mAb therapies and expectations derived therefrom

Patent Owner contends that "the art taught that many preventative migraine medications affected the CGRP signaling pathway." PO Resp. 40 (providing examples). Patent Owner then argues that "Petitioner fails to provide any explanation of why a POSA would have been motivated to treat a subject who has already failed at least two different classes of drugs that affect the CGRP pathway with yet another CGRP drug—galcanezumab." *Id.* (citing Ex. 2037 ¶ 227). We do not find this argument persuasive.

We agree with Dr. Hays that the POSA would have understood that "unlike traditional preventatives, anti-CGRP pathway mAbs 'targeted' CGRP in the sense that they specifically bind CGRP or the CGRP receptor."

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[19] Even if erenumab were understood to carry some advantages, that would not necessarily defeat Petitioner's obviousness challenge or the motivation for POSAs to have used alternative mAbs within the class. Suitable options may be obvious even where better options were known. *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021) ("[It is] not necessary to show that a combination is the best option, only that it be a suitable option.") (internal quotation marks and citation omitted).

Ex. 1239 ¶ 40; *see also* Tr. 79 (counsel for Patent Owner, explaining that "those molecules that were in the prior art were known to affect the CGRP pathway, even if they weren't doing it by directly binding to the ligand or by directly binding to the receptor").

We also agree with Dr. Hays that the "mechanistic distinction between anti-CGRP pathway mAbs and traditional migraine preventatives was recognized throughout the literature." Ex. 1239 ¶ 41 (citing evidence). Dr. Hays's testimony is consistent with prior art that discusses anti-CGRP treatments as having a distinct mechanism of action. Ex. 1230, 2 ("mAbs against CGRP or its receptor [CGRP mAbs] represent the unique disease-specific and mechanism-based migraine prevention treatment"); Ex. 1042, 1 ("Monoclonal antibodies antagonizing the CGRP pathway represent a novel approach to prevention: a mechanism-specific migraine-targeted therapy."); Ex. 1224, 7 (discussing the mAb, fremanezumab, as an "add-on therapy to other migraine prophylactics" and explaining that the idea is "plausible" because "all other migraine prophylactic medications act through non-CGRP mechanisms"); Ex. 1023, 7 (opining that mAbs against CGRP "deserve to be called a breakthrough" in part because they "were developed based on the pathophysiologic concept that the trigeminovascular system and CGRP have a key role in the development of migraine pain").

Perhaps in view of the mechanistic distinction between mAbs and other available treatments, the prior art discussed mAb treatments as novel and different from other types of migraine treatments. *See e.g.*, Ex. 1049, 1–2 ("CGRP antagonists showed efficacy in several clinical trials, but their severe side effects and adverse events with long-term administration discouraged further research. . . . The recently developed monoclonal antibodies against CGRP or its receptor (anti-CGRP mAbs) have triggered much interest in the headache community.”); Ex. 1025, 2–3 (“Although the small molecule agents that target the CGRP receptor are still under investigation, the recent development of humanized antibodies to CGRP and its receptor appear more promising for three important reasons: they are unlikely to cause liver toxicity or other serious AEs; they are biological products with extreme specificity for their target and very long half-lives, compared to oral medications; and they may have considerably better tolerability and safety profiles.”); Ex. 1053, 13 (“More-effective and better-tolerated acute and preventive treatments are needed for migraine and cluster headache patients in episodic, chronic and medically refractory subgroups. mAbs targeting CGRP or its receptor hold the most promise for preventive treatment in all of these subgroups”); Ex. 1028, 8 (“mAbs targeting the CGRP pathway are a promising new drug class that may provide a valuable new option for clinicians aiming to relieve the burden of individuals with episodic or chronic migraine”).

For the reasons discussed at length in her declaration (Ex. 1239 ¶¶ 43–62), we agree with Dr. Hays that the evidence cited by Patent Owner and Dr. Grosberg does not show that prior art non-mAb treatments “work against migraine in a manner analogous or equivalent to the anti-CGRP pathway mAbs” (*id.* ¶ 42). Considering this evidence together with the evidence that anti-CGRP mAbs were regarded as a novel class of drugs with a distinct mechanism of action, Patent Owner’s assertion that non-mAb prior art treatments also affect the CGRP pathway does not persuade us that the failure of such treatments materially detracts from the expectation that galcanezumab and erenumab would have similar efficacy or otherwise negatively impacts a POSA’s motivation to substitute one for the other.

(iv) Conclusion with respect to similar efficacy

Taking into consideration all of the arguments and evidence of record bearing on Petitioner's assertion that similar efficacy provides a motivation to combine –including, evidence of similar efficacy, evidence regarding CGRP pathway uncertainty, evidence regarding differences between galcanezumab and erenumab in mechanism of action, and evidence regarding the failure of other non-mAb treatments that affect the CGRP pathway – we find that Petitioner has established, by a preponderance of the evidence, that the POSA would have expected erenumab and galcanezumab to have similar efficacy. Patent Owner's evidence and argument to the contrary do not persuade us otherwise.

b) Sun's disclosure of combination therapy

Petitioner argues that the POSA would have been motivated to use galcanezumab in Sun's methods because Sun discloses "combination treatment with 'an anti-CGRP antibody . . . in . . . ***WO 2011/156324***'", which, according to Petitioner, disclosed galcanezumab. Pet. 30 (citing Ex. 1006 ¶ 250; Ex. 1101 ¶¶ 51, 145–147; Ex. 1043, 23). Petitioner contends that a POSA would "understand this to teach that the use of galcanezumab would be safe and effective in *Sun's* intended patients." *Id.* (citing Ex. 1101 ¶¶ 51, 149).

Patent Owner concedes that Sun "mentions galcanezumab" but argues that it does so "solely in the context of a hypothetical *combination* therapy with erenumab." PO Resp. 28. According to Patent Owner, Sun does not disclose a "therapeutically effective amount of galcanezumab administered alone or in combination," and does not disclose a "patient population for which a POSA would have expected success in administering galcanezumab." *Id.* Patent Owner cites the testimony of Dr. Grosberg, who testifies that combination therapy could be dangerous:

> Although antagonist antibodies against the CGRP pathway were safe in general migraine patients studied in the clinical trials of the prior art, a POSA would have expected that targeting the CGRP pathway at multiple points with different medication could have deleterious synergistic effects resulting in severe adverse events that targeting the CGRP pathway at a single point by a single antagonist antibody would not have.

Ex. 2037 ¶ 189. Dr. Grosberg also testifies that "common knowledge in the prior art suggest[ed] using medications having different mechanisms of action in combination therapy." *Id.* ¶ 190. Thus, Patent Owner asserts, absent data or explanation, Sun's mention of combination therapy provides the POSA "no motivation to combine galcanezumab with erenumab because of safety concerns and the fact that both drugs act at the *same point* (receptor-ligand interaction) on the *same pathway*." PO Resp. 29.

Although Sun discloses that its anti-CGRP receptor antibody (erenumab) "can be administered in combination with" galcanezumab (Ex. 1006 ¶ 250), Patent Owner and Dr. Grosberg raise what appear to be valid concerns regarding the use of galcanezumab and erenumab together in combination therapy. *See*, *e.g.,* Ex. 2037 ¶¶ 189–190. As Petitioner does not persuasively address these concerns, the record does not support that the POSA would have been motivated to use galcanezumab and erenumab together in combination therapy.

That said, Patent Owner's argument appears to be somewhat of a red herring. The Petition relies on Sun's disclosure of combination therapy as teaching that "the use of galcanezumab would be safe and effective in *Sun's* intended patients." Pet. 30 (citing Ex. 1101 ¶¶ 51, 149). As Petitioner's counsel clarified at oral argument, Petitioner does not propose actually using galcanezumab in combination with erenumab.

> I think Teva has misconstrued our argument a little bit. . . . We rely on Sun's teaching of combination therapy as a rationale for why a person of ordinary skill would understand that galcanezumab is entirely suitable, same with galcanezumab [sic, erenumab] in Sun's methods. . . . Our proposal is not to take erenumab from Sun and move it into Dodick. What is being argued is that Sun teaches, here are types of patients that can be useful -- usefully treated with CGRP antibodies. Galcanezumab has similar properties. Galcanezumab and fremanezumab would be obvious to use, those in the same patient populations. There's no need to consider whether a combination per se is included or excluded in Dodick, in our view.

Tr. 23. Thus, Petitioner contends only that the disclosure of combination therapy in Sun is evidence that galcanezumab would be safe and effective in Sun's patients. On this point, we agree with Petitioner. Ex. 1006 ¶ 250 (Sun disclosing treatment combination therapy with anti-CGRP antibodies known and described in WO 2011/156324); Ex. 1101 ¶¶ 51 (Dr. Evers testimony explaining that WO 2011/156324 discloses galcanezumab), 148 (Dr. Evers testimony that a POSA would understand Sun's disclosure as teaching that "galcanezumab would be effective in the patient populations it [Sun] disclosed").

c) Sun and Dodick's patient populations

Petitioner contends that "*Dodick's* eligibility criteria pointed to treating the same "RM" patients taught by *Sun.*" Pet. 30. More specifically, Petitioner contends that *Dodick* "allowed for treatment of patients who had failed two 'approved migraine preventative treatments'" and that "a POSA would have understood these patients to be the same as the 'failed' or 'intolerant' patients described by *Sun*." *Id.* According to Petitioner, this "would have motivated the substitution of galcanezumab—one of only four clinically proven anti-CGRP antibodies . . . — for *Sun's* erenumab." *Id.* at 31.

Patent Owner argues that Dodick "does not disclose . . . the refractory migraine patient population defined in the claims." PO Resp. 32. Patent Owner points out that Dodick's exclusion criteria preclude treatment of patients who failed to respond to "more than two adequately dosed approved migraine prevention treatments," which, Patent Owner notes, "expressly excludes most of the refractory migraine spectrum." *Id.* Patent Owner then argues that "[e]ven though Dodick does not exclude all of the claimed patient population (which includes two prior failures), it nonetheless makes clear that the study did not test the effectiveness of galcanezumab in refractory migraine patients." *Id.* at 32–33. Indeed, according to Patent Owner, "Dodick's exclusion criteria would have suggested to a POSA that refractory patients were *not* expected to be a successful treatment group, thus the reason for excluding them." *Id.* at 33.

We agree with Patent Owner that Dodick does not set forth inclusion criteria that require the presence of patients meeting the definition of "refractory" set forth in our claim construction. Ex. 1003, 2 (Dodick inclusion and exclusion criteria); *see also* Ex. 2037 ¶¶ 68–69 (Dr. Grosberg's testimony describing import of inclusion and exclusion criteria). Accordingly, Dodick, considered in isolation, does not speak to the effectiveness of galcanezumab in treating the claimed refractory patients.

We do not credit Patent Owner's argument that Dodick's exclusion criteria "would have suggested to a POSA that refractory patients were *not* expected to be a successful treatment group" because Patent Owner does not direct us to persuasive evidence supporting its argument. PO Resp. 33; *See Johnston v. IVAC Corp.*, 885 F.2d 1574, 1581 (Fed. Cir. 1989) ("Attorneys' argument is no substitute for evidence."); *In re Pearson*, 494 F.2d 1399, 1405 (CCPA 1974). Patent Owner cites paragraph 249 of Dr. Grosberg's declaration as support, but paragraph 249 does not support the proposition for which it was cited. Ex. 2037 ¶ 249. Instead, in the cited paragraph, Dr. Grosberg testifies that even if Group III refractory patients were included in Dodick's study, the POSA would not know how many patients were included or how they responded. *Id.* Moreover, Patent Owner's argument seems inconsistent with Dr. Grosberg's testimony that a POSA would have no expectation as to how an excluded patient would respond to treatment. *Id.* ¶ 69 ("A POSA would not have known how the patients with an exclusion criteria would respond to the medication studied in the clinical trial."); *see also* Ex. 1042, 8 ("An important group of patients, those that failed more than two preventive categories, was largely excluded from the trials, and thus, it is unknown what benefit this population would derive.").[20]

Consistent with our finding that the POSA would not conclude that treating excluded patients would be unsuccessful, we find that the absence of exclusion does not support an expectation that treating non-excluded patients would be successful. For this reason, we are not persuaded by Petitioner's argument that Dodick's eligibility criteria "pointed to" treating the claimed refractory patients.

[20] Even if we were to credit Patent Owner's argument that the POSA would draw an inference that excluded patients would not succeed, the impact of that inference would be limited because Dodick's exclusion criteria allow for the inclusion of at least some patients encompassed within the claimed refractory patient population – e.g., patients who failed two prior treatments.

d) Other arguments raised by Patent Owner

Patent Owner argues that the POSA "would not have been motivated to *change* the treatment described in patients for which Sun's erenumab was *successful*." PO Resp. 38. According to Patent Owner, "[a] POSA generally does not depart from or modify a method that is working as intended absent some specific teaching that doing so would satisfy an unmet need or overcome some identified deficiency." *Id.* 38. Conversely, Patent Owner argues that "nothing in Sun would have motivated a POSA to look to Dodick or Sharma to treat a patient who has *failed* Sun's method of erenumab treatment." *Id.* at 39; PO Sur-reply 8.

We do not find these arguments persuasive. If we were to accept Patent Owner's arguments at face value, there would be no need for an obviousness analysis, as most, if not all, obviousness challenges are based on a reference that discloses something that already works. Moreover, *KSR Int'l. Co. v. Teleflex Inc.,* instructs that "[i]f a person of ordinary skill can implement a predictable variation, § 103 likely bars its patentability. 550 U.S. 398, 421 (2007). As to Patent Owner's argument that the POSA would not have been motivated to treat a patient that failed Sun's method of treatment, this is little more than a strawman. Petitioner does not propose to use galcanezumab to treat patients that failed treatment with erenumab or any other mAb. Rather, Petitioner proposes that "[a] POSA would have been motivated to use galcanezumab in *Sun*'s methods." Pet. 30.

e) Conclusion regarding motivation to combine

We find that Petitioner has established, by a preponderance of the evidence that a POSA would have been motivated to use galcanezumab in place of erenumab in Sun's method. This finding is supported by:

1) evidence showing that galcanezumab and erenumab have similar efficacy, 2) evidence showing that mAbs, including galcanezumab and erenumab were treated as a class and recognized to have similar efficacy, and 3) evidence showing that prior art recognized both galcanezumab and erenumab, as well as mAbs in general, to have the potential to treat refractory migraine. *Aventis Pharma Deutschland GmbH v. Lupin, Ltd.*, 499 F.3d 1293, 1301 (Fed. Cir. 2007) (for motivation "it is sufficient to show . . . an expectation . . . [of] similar properties") (quotations omitted). Sun's disclosure of using galcanezumab in combination therapy provides further motivation to use galcanezumab in place of erenumab because it teaches that galcanezumab would be safe and effective in Sun's patients.

3. <u>Has Petitioner sufficiently supported that a POSA would have had a reasonable expectation of success in using galcanezumab to treat refractory migraine?</u>

Petitioner provides multiple reasons why the POSA would have expected success in treating refractory migraine patients. <u>**First**</u>, Petitioner argues that Dodick "demonstrated galcanezumab was effective in migraineurs" and an abstract from Headache, The Journal of Head and Face Pain, reports that galcanezumab was effective in "patients with a history of treatment failure." Pet. 35 (citing Ex. 1003, 1, 6–7; Ex. 1072, 74). <u>**Second**</u>, Petitioner contends that the success of other anti-CGRP mAbs – like erenumab and fremanezumab – support an expectation that galcanezumab, which similarly affects the CGRP pathway, would be similarly efficacious. *Id.* at 35–37. <u>**Third**</u>, according to Petitioner, anti-CGRP antibodies have a mechanism of action that is distinct from that of conventional non-mAb treatments such that a POSA would have expected an anti-CGRP mAb treatment, like galcanezumab, to "treat migraineurs independent of whether they used or failed conventional treatments that targeted non-CGRP pathways." *Id.* at 38. **Fourth**, Petitioner argues that plans for "numerous anti-CGRP mAbs clinical trials that allowed for inclusion of patients that failed two preventives further reinforced the reasonable expectation of success in treating the 'select[ed]' [refractory migraine] patients with galcanezumab.'" *Id.* at 39.

We discuss each of these four reasons why the POSA would expect success, and Patent Owner's arguments in response thereto, in turn.[21] The '161 patent sets a relatively low threshold for success in "treating or preventing migraine," defining that phrase to mean: "obtaining . . . ***improvement in any aspect*** of a refractory migraine, including lessening severity, alleviation of pain intensity, and other associated symptoms, reducing frequency of recurrence, reducing the number of monthly headache days or hours, increasing the quality of life of those suffering from refractory migraine, and decreasing dose of other medications (e.g., acute headache medication) required to treat the refractory migraine." Ex. 1002, 19:65–20:11 (emphasis added). Thus, in considering whether Petitioner has sufficiently supported that a POSA would have had a reasonable expectation of success, we consider whether the POSA would have expected improvement in any aspect of refractory migraine. Considering the evidence under this standard, we conclude that Petitioner has established, by a preponderance of the evidence, that the POSA would have expected galcanezumab to be successful in treating each of the three claimed patient groups.

[21] Although we discuss each of these positions individually, we do so purely for convenience, recognizing that expectation of success is determined based on consideration of the prior art as a whole.

a) Teachings in the art that galcanezumab was effective in treating migraine

Dodick and Sharma disclose Phase II studies showing the effectiveness of galcanezumab in the general migraine population. Ex. 1003, 6 (disclosing that galcanezumab was "effective and generally well tolerated for the preventive treatment of migraine in patients with frequent attacks" and "significantly reduced the mean number of migraine headache days over the 3-month study period"); Ex. 1007, 24 (disclosing that galcanezumab was "numerically superior to placebo on primary outcome measures at all-time points" with one dose arm (120 mg) providing "a significantly greater reduction compared to placebo in number of migraine headache days in the last 28 day period of the 12 week treatment phase"). And an abstract from *Headache*, The Journal of Head and Face Pain ("the Headache Abstract") teaches that galcanezumab was effective in patients with a "history of failure to preventative treatments." Ex. 1072, 74. Supported by the testimony of Dr. Evers, Petitioner contends that these teachings support that a POSA would reasonably expect "efficacy using galcanezumab in the claimed patients with prior treatment failures." Pet. 35; Ex. 1101 ¶ 166. Patent Owner presents arguments against each of Dodick, Sharma, and the Headache Abstract. We address each in turn.

(i) Arguments relating to Dodick

Patent Owner argues that "Dodick does not disclose the selection limitation or the refractory migraine patient population defined in the claims." PO Resp. 32. Rather, Patent Owner asserts, "Dodick reports a clinical trial studying galcanezumab . . . in non-refractory episodic migraine patients." *Id.* Moreover, Patent Owner argues, "Dodick . . . expressly excludes most of the refractory spectrum" suggesting that "refractory patients were *not* expected to be a successful treatment group." *Id.* at 32–33. As to the portion of the refractory spectrum that Dodick does not exclude, Patent Owner argues that Dodick "nonetheless makes clear that [its] study did not test the effectiveness of galcanezumab in the claimed refractory migraine patients." PO Sur-reply 17–18.

We agree with Patent Owner that Dodick, considered in isolation, does not speak to the effectiveness of galcanezumab in treating the claimed refractory patients. However, we do not consider Dodick in isolation; we consider it in connection with the knowledge of the POSA, which includes what is taught in the Headache Abstract, among other things. Likewise, we do not find Patent Owner's argument that Dodick fails to disclose the claimed patient population persuasive because Petitioner relies on Sun rather than Dodick for disclosure of the claimed patient population. Finally, for the reasons discussed *supra* § II.F.2.c, we do not credit Patent Owner's argument that Dodick's exclusion of "most of the refractory spectrum" would cause the POSA to expect failure.

(ii) Arguments relating to Sharma

Patent Owner argues that Petitioner's reliance on Sharma is misplaced because "Sharma provides only seven sentences of information regarding a trial of galcanezumab, without reporting any inclusion or exclusion criteria other than age and migraine frequency." PO Resp. 36. According to Patent Owner, "[b]ecause it is silent on prior treatment history, a POSA would not have been able to infer from Sharma any expectation of success in any subgroup of migraine patients (let alone the claimed group)." *Id.* This argument is similar to the arguments Patent Owner made with respect to Dodick, and we do not find it persuasive for the reasons already discussed.

Namely, we do not consider Sharma in isolation, and Petitioner relies on Sun rather than Dodick for disclosure of the claimed patient population.

(iii) Arguments relating to the Headache Abstract

Patent Owner argues that the Headache Abstract does not support an expectation of success because it relies on a *post-hoc* analysis. PO Resp. 49. Although a *post hoc* analysis of a subgroup of refractory patients may not be a strong as, for example, an analysis in which this patient subgroup was defined *a priori*, *post hoc* analyses were recognized and relied upon in the art. *See e.g.,* Ex. 1224, 7 ("post hoc analyses do play an important role in better understanding the benefits of any drug, including evaluation of subsets of patients that might experience particular benefit"). Moreover, the Federal Circuit has explained that "a person of skill in the present context *can* draw reasonable inferences about the likelihood of success even without a perfectly designed clinical trial showing a statistically significant difference in efficacy between a specific dose and placebo." *Acorda Therapeutics, Inc. v. Roxane Labs., Inc.*, 903 F.3d 1310, 1333–34 (Fed. Cir. 2018) ("This Court has long rejected a requirement of '[c]onclusive proof of efficacy' for obviousness.") (alteration in original). Here, although the analysis in the Headache Abstract was prepared *post hoc*, the variables analyzed were "pre-specified before study unblinding" and are described as "statistically significant." Ex. 1072, 73–74. Accordingly, although the Headache Abstract's may not be entitled to as much weight as a perfectly designed clinical trial designed to study galcanezumab in the claimed refractory patients, it may nonetheless help support an expectation that galcanezumab would be effective in treating patients with a "history of failure to preventative treatments."

Patent Owner argues that the Headache Abstract "reports on 'treatment effect' of the 'history of failure' in only 'the 300 mg dose group.'" PO Resp. 49. According to Patent Owner, "[t]his is a higher amount than the 120 or 275 mg galcanezumab dose suggested by Petitioner's other prior art." *Id.* at 49–50. We do not find this argument persuasive. Although Dodick discloses a 150 mg dose of galcanezumab (Ex. 1003, 1), Sharma discloses a 300 mg dose – i.e., the same dose tested in the Headache Abstract (Ex. 1007, 23 (disclosing doses of 5, 50, 120 and 300 mg and teaching that "all 4 dose arms were numerically superior to placebo on primary outcome measures at all-time points"). Moreover, claim 1 does not recite a particular dose.

Patent Owner argues that the Headache Abstract "contains no definition of 'history failure to preventive treatments,' no information regarding the quantify, type, or nature of 'failure,' and no number of patients with 'history of failure' who were actually treated with galcanezumab." PO Resp. 50. Patent Owner contends that "[w]ithout this information, a POSA would have no way to extrapolate from this study any expectations about the claimed patient population." *Id.*

We acknowledge that the Headache Abstract does not expressly define "history of failure to preventative treatments." Nonetheless, we do not agree with Patent Owner that this prevents the POSA from drawing inferences about the failed patients described in the Headache Abstract. For example, we find that the POSA would have inferred that the patients described in the Headache Abstract failed multiple treatments because: 1) the term "treatments" is plural, 2) the term "history" suggests more than one event, and 3) the evidence supports that the average migraineur has taken multiple prophylactic medications (Ex. 1015, 2 ("The mean number of prophylactic medications ever used was 2.92 for EM [episodic migraine] and 3.94 for CM [chronic migraine].").

Similarly, absent evidence to the contrary, the POSA would have inferred that the patients in the Headache Abstract had received typical preventative migraine treatments. Ex. 1101 ¶ 164 (Dr. Evers's testimony that the POSA "would have understood these results to be relevant to failure to the typical migraine treatments, such the Silberstein 2012 'Level A' and 'Level B' treatments as well as those in the '161 patent's 'clusters' and 'selecting' step"), *see also id.* ¶ 119 n.15 (Dr. Evers's testimony that the medication recited in the clusters in our claim construction "include the most commonly prescribed migraine prevention treatments and treatment classes" and that in his own practice, "it would be exceedingly rare to see a patient that had *not* had an 'inadequate response' to at least two different medications selected from the Patent's defined 'clusters.'").

Patent Owner argues that we should not credit Dr. Evers's testimony on the meaning of the term "refractory" because in his deposition, Dr. Evers "admitted that he does not understand the teachings of the very references he relied upon for alleged definitions of refractory." PO Resp. 12. It is not clear whether this argument applies to Dr. Evers's testimony on typical migraine treatments. *Id.* If it does, we are not persuaded that it undermines his credibility on this specific topic or, more generally, on how the term "refractory" was used in the art. In the testimony at issue, Dr. Evers testifies that he does not know what the authors of three references were referring to when they used the term "refractory." Ex. 2071, 70:15–17 ("I do not know what he [the author of Ex. 1013] is referring to by 'refractory migraine' in this context."), 72:2–6 ("I was not a member of this group [the American Headache Society], and I cannot tell you what they mean with 'refractory migraine' in this context."), 75:2–10 ("I cannot conclude from this text what Schuster is referring to, what is his definition for 'medically refractory subgroups.'"). This testimony does not call into question Dr. Evers's ability to speak to typical migraine treatments.

The evidence of record also supports that the POSA would have inferred that the Headache Abstract's failed patients underwent a treatment period of at least three months before being deemed to have failed treatment. As discussed *supra* § II.F.1.a.i, the evidence supports that three months was a very common minimum period for evaluating whether a treatment was successful. Absent a connection between the Headache Abstract and a reference suggesting a shorter evaluation period (like the tie between Sun and Tepper), we find that the POSA would have inferred a three-month evaluation period.[22]

As to the threshold for determining whether a treatment is successful, Dr. Evers testifies that "a preventative drug was considered clinically successful if it reduced migraine frequency and/or symptoms by at least 30% or 50%, depending on the criteria used." Ex. 1101 ¶ 23. To the extent Patent Owner's argument that Dr. Evers testimony is not credible applies to this topic (PO Resp. 12), we do not find it persuasive because it does not call into question Dr. Evers's ability to speak to the criteria for determining success. This is particularly true where Dr. Evers's testimony is consistent with the evidence he cites. *See* Ex. 1016, 4 ("Success is defined as a 50% reduction in attack frequency or headache days, a significant decrease in attack duration, or an improved response to acute medication."); Ex. 1017, 8

[22] Even if the POSA had inferred a shorter evaluation period, we find that it would not have impacted POSA's expectation of success for the reasons discussed *infra* § II.F.3.b.i.

("Responder rates should be defined as either ≥ 30% or ≥ 50% reduction in (i) headache days with moderate or severe intensity, (ii) migraine days, or (iii) migraine episodes compared with the baseline period. Responder rates have been traditionally defined in migraine as ≥ 50% reduction, but in CM population, a ≥ 30% responder rate can be clinically meaningful."). Thus, we find that a POSA would have inferred that the Headache Abstract's failed patients did not meet this criteria for success.

(iv) Conclusion with respect to teachings in the art about the effectiveness of galcanezumab

Dodick and Sharma support an expectation that galcanezumab would be effective in treating a broad genus of patients suffering from migraine. Ex. 1003, 3 (Dodick eligibility criteria), 6 ("[Galcanezumab] was effective and generally well tolerated for the preventative treatment of migraine in patients with frequent attacks."); Ex. 1007, 23 (teaching that galcanezumab was "numerically superior to placebo on primary outcome measures at all-time points" and that one dosage (120 mg) showed a "significantly greater reduction compared to placebo in the number of migraine headache days."). We recognize that neither Dodick nor Sharma discloses treating the claimed patient groups, but the Headache Abstract teaches that galcanezumab was effective in treating patients with a "history of failure to preventative treatments." Ex. 1072, 74 ("Compared with its respective counterpart, subgroups with statistically significantly greater treatment effect (galcanezumab vs placebo difference) at Month 3 included patients . . . [with a] history of failure to preventive treatments in the 300 mg dose group."). Accordingly, Dodick, Sharma and the Headache Abstract strongly support an expectation that galcanezumab would be successful in treating patient with a "history of failure to preventative treatments." For the reasons discussed above, we find that the POSA would have found patients with a "history of failure to preventative treatments" in the Headache Abstract to be quite similar to the claimed Group I patients. Thus, Dodick, Sharma and the Headache Abstract support an expectation that galcanezumab would be successful in treating the claimed Group I patients, particularly when considered with the additional evidence discussed *infra* §§ II.3.b–c. *See* Ex. 1101 ¶¶ 164–165 (Dr. Evers's credible testimony on expectation of success based on Dodick and the Headache Abstract).

b) Teachings in the art regarding the success of other anti-CGRP antibodies

Petitioner argues that the success of other anti-CGRP mAbs, including erenumab and fremanezumab helps to support an expectation of success.

As discussed above, Sun teaches that a monthly dose of 70 mg of the anti-CGRP antibody, erenumab, was "efficacious in preventing episodic migraine" and that efficacy was "similar regardless of . . . prior history of prophylactic medication use." Ex. 1006 ¶¶ 268, 272. In addition to these teachings, Sun also describes a prospective Phase II clinical trial and a prospective Phase III clinical trial. *Id.* ¶¶ 273–295. The results of the two prospective trials are described in separate, subsequent documents. Ex. 1037 ("Tepper," reporting on the Phase II trial); Ex. 1040 ("Goadsby," reporting on the Phase III trial). As to the Phase II trial, in which 49% of the patients "had failed at least two previous preventative drug classes," the results were described as suggesting that "there could b[e] efficacy in a treatment-resistant population." Ex. 1037, 8. The results of the planned Phase III trial, are similarly described as showing "[r]obust treatment effects . . . in subjects who had previously failed preventive migraine treatments . . . suggest[ing] that erenumab may have particular utility in this subgroup of patients." Ex. 1040, 13–14.

Although galcanezumab targets the CGRP ligand while erenumab targets the receptor, Petitioner argues that the POSA would expect the success of erenumab to extend to galcanezumab because the prior art teaches that targeting both the ligand and the receptor would be effective. Pet. 36 (citing Ex. 1034, which states: "The expectation is that . . . antibodies against both the ligand and receptor would prevent CGRP-induced activation of sensitized central trigeminal pathways, therefore decreasing headache frequency over time."). According to Petitioner, the expectation that galcanezumab would be as effective as erenumab was further supported by the "recognition that '***mAbs targeting CGRP or its receptor*** hold the most promise for preventive treatment in [episodic, chronic and ***medically refractory] subgroups***.'" *Id.* at 37 (quoting Ex. 1053, 13).

Relatedly, two studies of another anti-CGRP antibody, fremanezumab, show that fremanezumab was effective as a migraine prophylactic. Ex. 1038, 2 ("This study provides level 1b evidence (well conducted individual randomised controlled trial) that TEV-48125 is effective for the preventive treatment of chronic migraine"); Ex. 1039, 2 ("This study provides level 1b evidence (ie, a robust, randomized clinical trial) that two doses of TEV-48125, a monoclonal antibody against CGRP, are effective for the preventive treatment of high-frequency episodic migraine"); Ex. 1057, 6 (study characterized in Exhibit 1039); Ex. 1058, 5 (study characterized in Ex. 1038). A press release from Patent Owner ("the Teva Press Release") characterizes these two fremanezumab studies as showing an "impressive decrease in migraine days, amongst such highly refractory patients." Ex. 1041, 2.

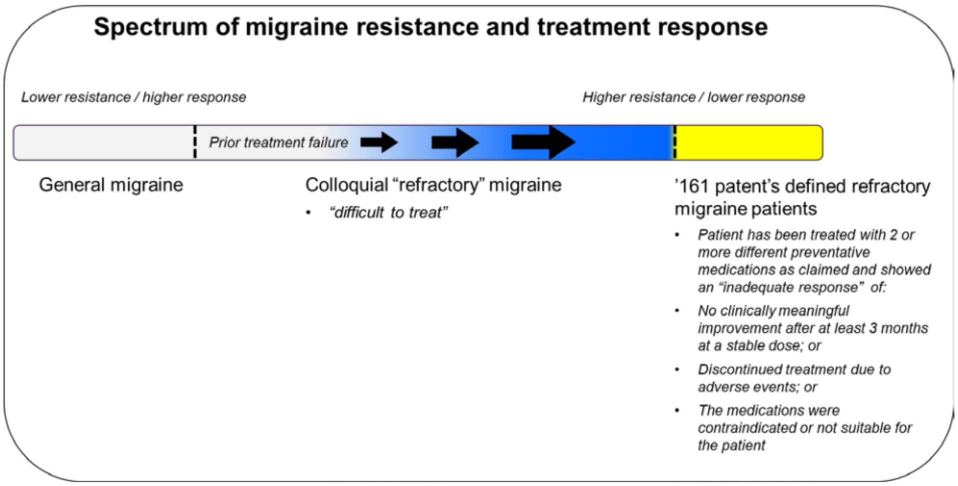
Petitioner argues that "the clinical success of erenumab and fremanezumab in patients with a history of treatment failure further provided a reasonable expectation that clinically successful galcanezumab—which operated by the same overall anti-CGRP mechanism and had demonstrated similar efficacy—would likewise be effective within the claimed method." Pet. 35. Patent Owner disputes that the alleged success of erenumab and fremanezumab supports an expectation of success.

Below, we address the arguments and evidence of record regarding the contributions of the clinical trials of erenumab and of fremanezumab to expectation of success as reflected in each of Sun, Tepper, Goadsby, and the fremanezumab clinical trials.

(i) Evidence that the POSA would have expected success based on Sun's Example 3

Sun's Example 3 discloses that a monthly dose of 70 mg of the anti-CGRP antibody, erenumab, was "efficacious in preventing episodic migraine" and that efficacy was "similar regardless of . . . prior history of prophylactic medication use." Ex. 1006 ¶¶ 268, 272. Petitioner argues that this helps to support an expectation of success. Pet. 35–36. Patent Owner argues that the results disclosed in Sun do not support an expectation of success because Sun does not disclose treating the claimed patients. More specifically, Patent Owner asserts that Sun does not support an expectation of success because the claimed Group I patients are more difficult to treat than Sun's inadequate response patients.

According to Patent Owner, "[a]t the time of the invention, migraine resistance to treatment was known to occur on a spectrum." PO Resp. 2. On this spectrum "more (or highly) 'refractory' migraine patients were known to be more difficult to treat than less (or partially) refractory migraine patients." *Id.* Patent Owner and Dr. Grosberg illustrate the spectrum in Figure 1, reproduced below.



Id. at 3; Ex. 2037 ¶ 63. Figure 1 depicts a "[s]pectrum of migraine resistance and treatment response" with "[l]ower resistance/higher response" on the left side of the spectrum and "[h]igher resistance/lower response" on the right side of the spectrum. *Id.* Figure 1 places "[g]eneral migraine" patients on the left end of the spectrum, indicating that they are the easiest to treat. *Id.* "Colloquial 'refractory' migraine" patients are the middle of the spectrum, and the "'161 patent's defined 'refractory migraine'" patients are on the right end of the spectrum, indicating that they are the most difficult to treat. *Id.*

Patent Owner argues that "[t]he selected inadequate response patients of the '161 patent are more resistant to treatment and more prone to treatment failure than other migraine patients in the prior art (including other so-called refractory patients)." PO Resp. 4. More specifically, Patent

Owner identifies two differences that it contends differentiate Sun's "inadequate response" patients from the claimed Group I patients. First, Patent Owner contends that Sun's results do not "distinguish[] the number of prior failures," arguing that "[p]atients who fail a *single* prior treatment are not refractory migraine patients, and have a different expectation of success than patients who have demonstrated a *pattern* of failure (as defined in the '161 patent)." PO Resp. 42.[23] Second, Patent Owner also argues that "a study disclosing successful treatment in patients who failed a prior treatment after six weeks at a stable dose," as we found Sun to disclose, "does not give rise to a reasonable expectation of success among patients who failed a prior treatment after three months at a stable dose." *Id.* at 47–48. We do not find these arguments persuasive.

The record includes evidence supporting that with anti-CGRP mAbs, failure of prior preventative migraine prophylactics is not predictive of how hard patients are to treat with anti-CGRP mAbs. For example, contrary to the expectation that they would be harder to treat, the Headache Abstract teaches that patients who failed to respond to prior preventative treatments showed a "statistically greater treatment effect" than their counterparts when administered 300 mg of galcanezumab. Ex. 1072, 74. From this, the Headache Abstract concludes that failure to respond to preventative treatments may be a "predictor[] of clinical response with *greater* treatment effect." *Id.* (emphasis added). Similarly, Goadsby teaches that treatment of

[23] The evidence Patent Owner cites to support this statement (Ex. 2037 ¶¶ 192–195) does not address the difference between where on the spectrum of resistance patients who fail a single treatment and patients who fail two or more treatments. As Patent Owner does not elsewhere direct us to evidence supporting this contention, Patent Owner's position with respect to number of treatments is thus nothing more than attorney argument.

patients who failed prior treatments with 140 mg of erenumab had *greater* efficacy than the treatments in the overall trial population. Ex. 1040, 14–15 (“Robust treatment effects were observed for both 70 mg and 140 mg erenumab in subjects who had previously failed preventative migraine treatments. For 140 mg, effects were numerically greater in this subpopulation than in the overall trial population.”). And Sun itself reports that the efficacy of erenumab “was similar regardless of . . . prior history of prophylactic medication use.” Ex. 1006 ¶ 268, Fig. 7B.[24] These teachings are consistent with the evidence that anti-CGRP mAbs work by a different mechanism of action than other migraine prophylactics, and thus might be expected to succeed where other treatments had failed. *See infra* § II.F.3.b.iii.

We weigh this evidence supporting that failure of prior treatment is not predictive of how hard patients are to treat with anti-CGRP antibodies against Dr. Grosberg’s testimony – the only evidence offered by Patent Owner to support that Group I patients are more difficult to treat than the cited prior art patients. Dr. Grosberg testifies broadly that “a POSA would have concluded that the prior-art patients with migraine characterized as simply refractory could be on a lower end of the resistance spectrum and were easier to treat than the defined refractory migraine patients of the ’161 patent.” Ex. 2037 ¶ 59. With respect to length of treatment, Dr. Grosberg asserts that patients who fail treatment after three months fall on the more resistant end of the spectrum of resistance to treatment than patients who fail treatment after only six weeks. *Id.* ¶ 59 (“A POSA would have known that a

[24] We address Patent Owner’s argument that Sun and Goadsby lack statistical significance later in this section and *infra* § II.F.3.b.iii respectively.

patient who did not show clinically meaningful improvement after a longer evaluation period with a medication was more resistant to treatment than a patient who did not show improvement after a shorter evaluation period with the medication."). Dr. Grosberg explains: "This is because, for example, a patient who did not show improvement in a shorter evaluation period (e.g., 6 weeks) could still show improvement with the same medication after treatment for a longer period (e.g., 3 months) and perhaps was not even resistant to treatment in the first place." *Id.* According to Dr. Grosberg, "[a] POSA would have known that such patients (with no clinically meaningful improvement in migraine after less than three months of prior preventive medication therapy) were easier to treat than the Group I patients of the defined refractory migraine patients." *Id.* (emphasis omitted)

We recognize that, logically speaking, requiring a patient to fail more treatments and giving a patient more time to respond to treatment before designating that treatment a failure will likely narrow the subset of patients deemed refractory. *Id.* ¶ 59 (Dr. Grosberg's testimony making this point with respect to length of treatment); *see also* Ex. 1013, 4 (recognizing that an evaluation period longer than 2 months "would be preferrable," but concluding that such an evaluation period would "prolong the time necessary to meet refractory criteria" and thus "prevent patients receiving the appropriate level of care."). However, Dr. Grosberg does not cite to any evidence to support a correlation between the time it takes to respond to treatment with one medication, or the number of prior treatments a patient has failed, and the likelihood that treatment with an anti-CGRP mAb that acts by a different mechanism of action will be successful. Ex. 2037 ¶ 59 (Dr. Grosberg's testimony lacking citation to evidence); *see also infra* § II.F.3.c (discussing mechanism of action of anti-CGRP mAbs as compared to other migraine treatments). Indeed, more generally, none of Dr. Grosberg's multiple invocations of a spectrum of resistance cite any evidence to support that such a spectrum was recognized in the art. *See* Ex. 2037 ¶¶ 59, 63, 64, 82, 129, 140, 141, 184, 205, 224, 236, 254, 267, 275 (discussing spectrum of evidence without citation to evidentiary support). This significantly diminishes the persuasiveness of Dr. Grosberg's testimony. *Xerox Corp. v. Bytemark, Inc.*, IPR2022-00624, Paper 9 at 15 (PTAB Aug. 24, 2022) (precedential) (finding that conclusory testimony that "does not cite to any additional supporting evidence or provide any technical reasoning to support [that testimony] . . . is entitled to little weight"); *see also* 37 C.F.R. § 42.65(a) ("Expert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.").

Patent Owner argues that Dr. Grosberg's spectrum of resistance testimony is supported by the use of adjectives to modify the word "refractory" in the prior art. PO Resp. 45–46. More specifically, Patent Owner argues that the uses of "partially" and "highly" to modify the word "refractory" in two articles "confirms that 'refractory' conditions (in a general sense) exist on a spectrum along which treatment success (and therefor expectations of success) varies." *Id.; see also* Tr. 36 (counsel for Patent Owner arguing that their use of "modifiers like the words 'partially' and 'highly refractory' denote a spectrum of refractory or refractoriness"). We agree that the use of adjectives to modify the term "refractory" lends support to the notion that there may be degrees of refractoriness. Ex. 1041, 2 ("The very fast onset of preventative response, seen after a single dose of therapy, along with the impressive decrease in migraine days, amongst such ***highly refractory patients***, may bring us a step closer to provide widespread relief to people who suffer from chronic and episodic migraine.") (emphasis added); Ex. 1042, 8 ("Patients seen at specialty headache centers are often medically ***partially refractory***, having failed a number of preventives with limited remaining options.") (emphasis added). But such support is weak. And Dr. Grosberg does not cite these articles. More importantly, neither of these articles speaks to whether the length or number of treatments impacts degree of refractoriness. Nonetheless, we give these articles some weight as supporting that there are degrees of refractoriness.

In considering the evidence regarding Dr. Grosberg's spectrum of resistance as it relates to length of treatment, we also give some weight to the evidence supporting that the evaluation period for determining whether a treatment is effective is not critical. Ex. 1196, 3 ("Although there is no evidence on the optimal length of prophylactic treatments, 3 months is usually considered sufficient to assess prophylactic efficacy,"); Ex. 1234, 198:18–199:13 (testimony of Dr. Grosberg: "I agree [with the statement in Ex. 1196] that there was no evidence in the optimal length of prophylactic treatments at the time").

Considering all of the evidence with respect to the impact of the treatment evaluation period, we are not persuaded that the difference between Sun's 6-week evaluation period and the claimed 3-month evaluation period would have a material impact on a POSA's expectation of success. Similarly, we are not persuaded that the difference between Sun's unspecified number of treatment failures (*see* Ex. 1006 ¶ 268, Fig. 7B) and the claimed inadequate response to at least two preventative medications would have a material impact on a POSA's expectation of success. We give some weight to Dr. Grosberg's testimony to the contrary (*see*, *e.g.,* Ex. 2037 ¶ 59) but find that this testimony, even with the modest support lent by the use of adjectives "highly" and "partially" to modify "refractory," is outweighed by the evidence supporting that patients who had failed prior treatments responded to anti-CGRP treatment as well as, if not better, than patients who had not failed prior treatments. With respect to length of treatment, the evidence that there is no recognized optimal evaluation period further supports our conclusion.

Having determined that the length and number of treatments does not materially impact a POSA's expectation of success, we are not persuaded that the POSA would have viewed the claimed narrower subset as more difficult to treat with anti-CGRP mAbs than patients, like those disclosed in Sun, who fail following at least 6 weeks of treatment with an unspecified number of treatments. Put another way, the evidence of record does not support that a POSA would have different expectations for how, for example, a patient who failed to respond to six weeks of treatment with a beta blocker would respond to treatment with an anti-CGRP mAb than for a patient who failed to respond to three months of treatment with a beta blocker and an anti-convulsant.

Patent Owner argues that the data in Sun's Figure 7B, which supports Sun's teaching that the efficacy of 70 mg of erenumab, was "similar regardless of . . . prior history of prophylactic medication use" (Ex. 1006 ¶ 268), "are not statistically significant over placebo." PO Resp. 25–26. As support, Patent Owner cites the testimony of Dr. Grosberg, who testifies that based on the p-value in Figure 7B, a POSA would have understood that "the data is not strong enough to suggest that erenumab was effective in patients with previous treatment history." Ex. 2037 ¶ 109. Dr. Grosberg's testimony that the POSA would not accord Figure 7B statistical significance is unrebutted and credible. This diminishes the weight we accord to Sun's teaching that the efficacy of 70 mg of erenumab, was "similar regardless of . . . prior history of prophylactic medication use."

However, the weight we give Sun's teaching on efficacy is bolstered, to some extent, by that fact that it is consistent with other, similar statements in the art supporting that prior treatment failures may not be predictive of whether treatment with an anti-CGRP mAb will be successful. *See*, *e.g.*, Ex. 1072, 74 (discussed *supra* § II.F.3.a.iii); Ex. 1037 (discussed *infra* § II.F.3.b.ii); Ex. 1040, 16 (discussed *infra* § II.F.3.b.iii); and Ex. 1040, 329 ("Erenumab 140 mg showed better efficacy in patients who had failed ≥1 or ≥ 2 prophylactic medications.").[25] It is also bolstered, to some extent, by the evidence that anti-CGRP mAbs act through a different mechanism of action than conventional treatments. *See infra* § II.F3.b.iii. Overall, we find that Sun's statement still carriers some weight even without data supporting its statistical significance. *See Acorda Therapeutics,* 903 F.3d at 1333 (Fed. Cir. 2018) (find that a POSA "can draw reasonable inferences about the likelihood of success even without a perfectly designed clinical trial showing a statistically significant difference in efficacy").

In sum, we find that Sun's Example 3, which teaches that the efficacy of 70 mg of erenumab, was "similar regardless of . . . prior history of prophylactic medication use" (Ex. 1006 ¶¶ 268, 272), provides some support for the expectation that erenumab would be successful in treating the

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[25] Exhibit 1040 is entitled Abstracts of the 18th International Headache Congress. Ex. 1040, 1. It is comprised of multiple abstracts, including the abstract referred to herein as "Goadsby." *Id.* at 15–16. The citation to page 329 of Exhibit 1040 is not to Goadsby, but to a different abstract within the same collection of abstracts. It appears the non-Goadsby abstract relates to the same clinical trial as is described in Tepper. *Compare id.* at 328-329, *with* Ex. 1037.

claimed patient groups. This, in turn, helps to support an expectation that galcanezumab would be successful in treating the claimed patient groups. *See* Ex. 1101 ¶¶ 166–174 (Dr. Evers's credible testimony that a POSA would have reasonably expected that "*Sun*'s results for its receptor mAb would translate to *Dodick*'s ligand-targeting anti-CGRP mAb.").

(ii) Evidence that the POSA would have expected success based on the results of Sun's proposed Phase II clinical as reported in Tepper

Sun describes a prospective Phase II clinical trial of erenumab, the results of which are described in a subsequently published reference, Tepper. Ex. 1006 ¶¶ 273–289; Ex. 1037; PO Resp. 50 (tying Sun's prospective clinical trials to Tepper and Goadsby). The published results show that, in the Phase II trial, "453 patients (68%) . . . had failed at least one previous preventive drug class because of lack of efficacy or poor tolerability and 327 (49%) . . . had failed at least two previous drug classes." Ex. 1037, 8. Tepper describes the results of the trial as suggesting that "there could b[e] efficacy in a treatment-resistant population." *Id.* Petitioner argues that this helps to support an expectation of success. Pet. 35–36.

Patent Owner argues that Tepper is a *post hoc* analysis in which "only 185 patients [who had] 'previous preventive treatment failure[s]' []due to 'lack of efficacy or poor tolerability' were actually treated with erenumab, even though 327 total such patients were enrolled in the trial." PO Resp. 50. Patent Owner also argues that because Tepper does not define "lack of efficacy" or "poor tolerability" "a POSA would not have understood those terms to disclose the claimed inadequate responses." *Id.* According to Patent Owner, there is also "no indication of what classes prior treatments fell into, or any suggestion that [Tepper] treated even a single patient who failed *two* prior preventive treatments from different classes." *Id.* at 51. Finally, Patent Owner argues that Tepper "explicitly indicates that patients were evaluated for only 'six weeks' to determine prior failure of preventive medication, which a POSA would have understood to be a more lenient definition of prior 'failure.'" *Id.* We do not find these arguments persuasive.

As we discussed *supra* § II.F.3.a.iii, *post hoc* analyses were recognized and relied upon in the art, even though they may not be a strong as, for example, an analysis in which such patients were defined *a priori*. *See, e.g.*, Ex. 1224, 7. As also explained *supra* § II.F.3.a.iii "a person of skill . . . *can* draw reasonable inferences about the likelihood of success even without a perfectly designed clinical trial showing a statistically significant difference in efficacy between a specific dose and placebo." *Acorda Therapeutics*, 903 F.3d at 1333–34. Here, although Tepper's analysis may not be entitled to as much weight as a perfectly designed clinical trial designed to study erenumab in the claimed refractory patients, it nonetheless helps support an expectation that erenumab would be effective in treating patients with a "history of failure to preventative treatments." Similarly, although Patent Owner criticizes the sample size as including "only 185 patients," the sample size was sufficient to "suggest[]" to the authors that "there could b[e] efficacy in a treatment-resistant population." Ex. 1037, 8. While a larger sample would likely carry greater weight, we find that a sample of 185 patients was sufficient to allow the POSA to draw reasonable inferences about the likelihood of success.

Patent Owner's argument that Tepper "explicitly indicates that patients were evaluated for only 'six weeks' to determine prior failure of preventive medication," is not persuasive. For the reasons discussed *supra* § II.F.3.b.i, we are not persuaded that the difference between a 6-week evaluation period and the claimed 3-month evaluation period would have a material impact on a POSA's expectation of success.

As to Patent Owner's argument that Tepper does not define "lack of efficacy," Tepper states that "[r]eductions of headache frequency from baseline of more than 30% and by more than one day per month are generally thought to represent a clinically relevant change." Ex. 1037, 8. This is consistent with Dr. Evers's testimony that "a preventative drug was considered clinically successful if it reduced migraine frequency and/or symptoms by at least 30% or 50%, depending on the criteria used." Ex. 1101 ¶ 23; *see also* Ex. 1016, 4 ("Success is defined as a 50% reduction in attack frequency or headache days, a significant decrease in attack duration, or an improved response to acute medication."); Ex. 1017, 8 ("Responder rates should be defined as either ≥ 30% or ≥ 50% reduction in (i) headache days with moderate or severe intensity, (ii) migraine days, or (iii) migraine episodes compared with the baseline period. Responder rates have been traditionally defined in migraine as ≥ 50% reduction, but in CM population, a ≥ 30% responder rate can be clinically meaningful."). We find that a POSA would have inferred that patients who failed for "lack of efficacy" failed to meet this criteria for success.

As to Patent Owner's argument that Tepper does not define "poor tolerability," Tepper uses the term to describe a reason why patients failed treatment. Ex. 1037, 8 ("This study provided robust representation of . . . subgroups of patients . . . who had failed at least one previous preventive drug class because of . . . poor tolerability."); *see also id.* at 2 ("Oral preventive therapies available at present . . . are often . . . poorly tolerated, which can lead to low adherence rates.") (footnote omitted). Such patients are sufficiently similar to the claimed Group II patients (i.e., patients for whom adverse events made continued treatment intolerable) that Tepper helps to support an expectation of success in the claimed Group II patients.

Finally, we are not persuaded by Patent Owner's argument that Tepper provides "no indication of what classes prior treatments fell into, or any suggestion that [Tepper] treated even a single patient who failed two prior preventive treatments from different classes." Tepper expressly discloses patients who failed two drug classes. Ex. 1037, 8 (disclosing that 327 patients (49% of the total) "had failed at least two preventative drug classes."). And, as Petitioner points out, and Patent Owner does not dispute, the preventative classes of prior preventatives in this Phase II trial largely mirror those of the claims. Pet. Reply 17; *compare* Ex. 1223, 2 (listing prior preventives from Phase II study discussed in Ex. 1037), *with* Ex. 1002, 19:27–35 (listing classes of preventatives incorporated in claim construction). Moreover, even if the classes of treatments where not enumerated, a POSA would have understood "preventives" to include typical preventative migraine treatments. Ex. 1101 ¶¶ 116, 119 (discussed *supra* § II.F.3.a.iii).

In sum, we find that Tepper provides strong support for the expectation that erenumab would be successful in treating the claimed patient groups. This, in turn, helps to support an expectation that galcanezumab would be successful in treating the claimed patients groups. *See id.* ¶¶ 168–175 (Dr. Evers's credible testimony that a POSA would have reasonably expected that "*Sun*'s results for its receptor mAb would translate to *Dodick*'s ligand-targeting anti-CGRP mAb.").

(iii) Evidence that the POSA would have expected success based on the results of Sun's proposed Phase III clinical as reported in Goadsby

Sun describes a prospective Phase III clinical trial of erenumab, the results of which are described in a subsequently published reference, Goadsby. Ex. 2006 ¶¶ 290–295; Ex. 1040, 15–16; PO Resp. 50 (tying Sun's prospective clinical trials to Tepper and Goadsby). Goadsby describes the results of the Phase III trial as showing "[r]obust treatment effects . . . in subjects who had previously failed preventive migraine treatments . . . suggest[ing] that erenumab may have particular utility in this subgroup of patients." Ex. 1040, 13–14. Petitioner argues that this helps to support an expectation of success. Pet. 35–36.

Patent Owner's arguments with respect to Goadsby are similar to its arguments for Tepper. Patent Owner argues that Goadsby provides only a *post hoc* data analysis. PO Resp. 51. We do not find this persuasive because, as discussed *supra* §§ II.F.3.a.iii and II.F.3.b.ii, the law does not require "'[c]onclusive proof of efficacy' for obviousness" and *post hoc* analyses are recognized and relied upon in the art. Ex. 1224, 7; *Acorda Therapeutics*, 903 F.3d at 1333–34.

Patent Owner argues that Goadsby "does not differentiate results between subjects who had 'failed' only one prior preventive treatment and subjects who had failed two or more prior preventive treatments (let alone two or more treatments from different clusters), preventing a POSA from assessing effects separately in these groups." PO Resp. 51. This argument appears to be misplaced, as Goadsby's Table 1, reproduced below, provides data broken down by category.

Table 1. Change from Baseline in MMD and ≥ 50% Responder Rate

	PLACEBO			ERENUMAB 70MG			ERENUMAB 140MG		
	Overall[a] n=316	Failed ≥1 n=126	Failed ≥2 n=54	Overall[a] n=312	Failed ≥1 n=127	Failed ≥2 n=49	Overall[a] n=318	Failed ≥1 n=116	Failed ≥2 n=58
MMD, LS mean (SE)	-1.8 (0.2)	-0.6 (0.4)	-0.2 (0.8)	-3.2 (0.2)	-2.6 (0.4)	-1.6 (0.7)	-3.7 (0.2)	-3.2 (0.4)	-3.0 (0.7)
Difference from placebo (95% CI)	-	-	-	-1.4 (-1.9,-0.9)	-2.0 (-2.8, -1.2)	-1.3 (-2.6, 0.0)	-1.9 (-2.3,-1.4)	-2.5 (-3.4, -1.7)	-2.7 (-4.0, -1.4)
p value	-	-	-	*p*<0.001*	*p*<0.001	*p*=0.051	*p*<0.001*	*p*<0.001	*p*<0.001
≥ 50% responder rate[b]	26.6	17.5	11.1	43.3	38.6	26.5	50.0	39.7	36.2
Difference from placebo (%)	-	-	-	16.7	21.1	15.4	23.4	22.2	25.1
Odds ratio (95% CI)[c]	-	-	-	2.1 (1.5, 3.0)	2.9 (1.6, 5.3)	2.9 (1.0, 8.3)	2.8 (2.0, 3.9)	3.1 (1.7, 5.5)	4.5 (1.7, 12.4)
p value	-	-	-	*p*<0.001*	*p*<0.001	*p*=0.045	*p*<0.001*	*p*<0.001	*p*=0.002

CI=confidence interval; LS=least squares; MMD=monthly migraine days. [a]Overall STRIVE trial population. [b]≥50% responder rate is a ≥50% reduction from baseline in MMDs. [c]Odds ratio for erenumab vs placebo. *Statistically significant from placebo in the overall population. Statistical significance was not assessed in the subgroup analyses.

Table 1, is entitled "Change from Baseline MMD and ≥ 50 Responder Rate." It provides data on mean monthly migraine days ("MMD"), and difference from placebo and 95% confidence interval ("CI") for both tested doses of erenumab (70 mg and 140 mg) broken down into three patient groups: all patients, failed ≥ 1, and failed ≥ 2. It also provides data on ≥ 50% responder rate, % difference from placebo, and odds ratio and 95% confidence interval broken down into three patient groups: all patients, failed ≥ 1, and failed ≥ 2. Accordingly, Goadsby does differentiate results between subjects.

We recognize that Table 1 indicates that "[s]tatistical significance was not assessed in the subgroup analysis." Patent Owner correctly identifies this as a weakness in Goadsby's data. PO Resp. 28. This diminishes the weight we accord to Goadsby's teaching that galcanezumab showed "[r]obust treatment effects . . . in subjects who had previously failed preventive migraine treatments." Ex. 1040, 15. However, Goadsby interpreted its own data as "suggest[ing] that erenumab may have particular utility in this subgroup of patients." *Id.* at 16. In addition, the weight we give Goadsby's teaching on efficacy is bolstered, to some extent, by that fact that it is consistent with other, similar statements in the art. *See, e.g.,* Ex. 1072, 74 (discussed *supra* § II.F.3.a.iii); Ex. 1006 ¶ 268 (discussed *supra* § II.F.3.b.i); Ex. 1037 (discussed *infra* § II.F.b.ii); Ex. 1040, 329 ("Erenumab 140 mg showed better efficacy in patients who had failed ≥1 or ≥ 2 prophylactic medications.").[26] Overall, Goadsby's statement still carriers some weight even if the data is not identified as having statistical significance. *Acorda Therapeutics,* 903 F.3d at 1333.

Patent Owner argues that a post filing date reference teaches that "[o]nly 172 patients who had a 'history of preventive treatment failure' due to 'lack of efficacy' and only 127 due to 'unacceptable side effects' were actually treated with erenumab." PO Resp. 51. We do not consider this disclosure relevant because it would not have been available to the POSA at the time of the invention. Even if we were to consider this post filing evidence, we would not find it persuasive at least because Patent Owner does not direct us to persuasive evidence that the POSA would have considered such sample sizes insufficient. And the authors of Goadsby found the sample size sufficient to suggest that "erenumab may have particular utility in this group of patients." Ex. 1040, 16. Accordingly, we do not see the post-filing disclosure of the sample sizes in Goadsby as helpful to Patent Owner.

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[26] As noted above, it appears this abstract relates to the same clinical trial as is described in Tepper. *Compare* Ex. 1040, 328-329 *with* Ex. 1037.

Patent Owner argues that Goadsby does not provide a "definition of 'failure' (such as the duration of treatment employed)" and thus does not support an expectation of success. PO Resp. 51. To support its argument, Patent Owner directs us to post-filing evidence that, in Goadsby, failure was assessed after only six weeks. *Id.* at 51 (citing Exs. 2073 and 2074). Patent Owner's argument is unavailing.

For the reasons discussed *supra* § II.F.3.b.i, we are not persuaded that the difference between a 6-week evaluation period and the claimed 3-month evaluation period would have a material impact on a POSA's expectation of success. Thus, even considering the six-week treatment length disclosed in Patent Owner's post-filing evidence does not materially impact a POSA's expectation of success.

As to the other aspects of how "failure" is defined, in particular, the threshold for determining whether a treatment is successful, for the reasons discussed *supra* § II.F.3.a.iii, we find that a POSA would have considered a drug to have failed if it failed to reduce migraine frequency and/or symptoms by at least 30% or 50%. *See* Ex. 1101 ¶ 23; Ex. 1016, 4; Ex. 1017, 8. This is very similar to the threshold specified in our claim construction and thus the absence of an express definition in Goadsby does not detract from expectation of success.

The post-filing evidence Patent Owner directs us to indicates that failure was defined as "no reduction in headache frequency, duration, or severity after administration of the medication for ≥6 weeks at the generally accepted therapeutic dose(s) based on the investigator's assessment." Ex. 2073, 4 (study exclusion criteria, defining "[n]o therapeutic response"). This definition encompasses patients who, if we were to credit Dr. Goadsby's spectrum of resistance, are at least as refractory as the claimed patients.[27] Accordingly, even considering the post-filing evidence of Goadsby's threshold for success, Goadsby's definition of failure does not detract from a POSA's expectation of success.

In sum, we find that Goadsby provides some modest support for the expectation that erenumab would be successful in treating the claimed patient groups. This, in turn, helps to support an expectation that galcanezumab would be successful in treating the claimed patients groups. *See* Ex. 1101 ¶¶ 167–174 (Dr. Evers's credible testimony that a POSA would have reasonably expected that "*Sun*'s results for its receptor mAb would translate to *Dodick*'s ligand-targeting anti-CGRP mAb.").

(iv) Evidence that the POSA would have expected success based on clinical trials of fremanezumab

Two studies of another anti-CGRP antibody, fremanezumab, show that fremanezumab was effective as a migraine prophylactic. Ex. 1038, 2 ("This study provides level 1b evidence (well conducted individual randomised controlled trial) that TEV-48125 [fremanezumab] is effective for the preventive treatment of chronic migraine"); Ex. 1039, 2 ("This study provides level 1b evidence (ie, a robust, randomized clinical trial) that two doses of TEV-48125, a monoclonal antibody against CGRP, are effective for the preventive treatment of high-frequency episodic migraine"); Ex. 1057, 6 (study characterized in Exhibit 1039); Ex. 1058, 5 (study characterized in

[27] Arguably, Goadsby encompasses patients who are more refractory because Goadsby required no reduction in frequency, duration, or severity, while the claim requires an absence of "clinically meaningful improvement." *Id.* ¶ 65; Ex. 1234, 73:7–75:13 (explaining that the most refractory patients on the spectrum of resistance have the lowest response to treatment).

Ex. 1038). A press release from Patent Owner ("the Teva Press Release") characterizes these two fremanezumab studies as follows:

> "The collective data generated from these studies herald promise for millions of people who suffer from episodic and chronic migraines, a disease with substantial implications and unmet needs," stated Marcelo E. Bigal, Teva's Head of Global Clinical Development for Migraine and Headaches [and a named inventor on the '161 patent]. "The very fast onset of preventive response, seen after a single dose of therapy, along with the impressive decrease in migraine days, ***amongst such highly refractory patients***, may bring us a step closer to provide widespread relief to people who suffer from chronic and episodic migraine."

Ex. 1041, 2 (emphasis added). Petitioner argues that these studies help to support a reasonable expectation of success. Pet. 36–37.

Patent Owner argues that although the Teva Press Release uses the phrase "highly refractory patients," the clinical trials on which the Teva Press Release was based "do not recite the selected refractory migraine patients of the claimed method in their inclusion criteria," were not "designed to study such patients," and do not "provide any data or outcomes for such patients." PO Resp. 52. Thus, according to Patent Owner, "a POSA would not have understood the phrase 'highly refractory patients' in Teva Press Release . . . to mean the refractory migraine subjects of the claimed method." *Id.* Instead, Patent Owner asserts, "a POSA would have, at most, understood these phrases to mean a nonspecific resistan[ce] to treatment." *Id.* (citing Ex. 1053 and Ex. 1059).[28]

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[28] Patent Owner makes the same argument for the use of the term "relatively refractory" in Exhibit 1023. PO Resp. 52. We do not find this argument for the reasons discussed in connection in connection with the Teva Press Release.

We are not persuaded that differences in how the term "refractory" is used in the Teva Press Release and in the '161 patent adversely impact the expectations of success a POSA would have based on the success of fremanezumab. The present record tends to support that the term "refractory" was understood in the art to have a definition similar to and, in some respects, more restrictive than, that recited in the '161 patent. *See* Ex. 1101 ¶¶ 20–25 (testimony of Dr. Evers that refractory migraine was "generally recognized to describe patients that failed at least three drugs"); Ex. 1012, 13 ("There is no agreement on the number of drugs a patient should have received before being considered refractory, but it is common consensus that at least three or four drugs belonging to the four most effective pharmacological classes (beta blockers, anticonvulsants, calcium antagonists, tricyclic antidepressants) should have been adequately tested."); Ex. 1011, 3, table 1 (summarizing various approaches to defining "refractory," including requiring "failure of at least 4 classes" of treatments, failure "from at least 2 of 4 drug classes," and failure of "[t]he greatest possible number of drugs"); Ex. 1013, 1 (proposing to define "refractory migraine" to require that "patients fail adequate trials of preventative medicines, alone or in combination, from at least 2 to 4 drug classes . . . [p]atients must also fail adequate trials of abortive medicines . . . and *either* nonsteroidal anti-inflammatory drugs (NDAISs) *or* combination analgesic"); Ex. 1014, 509 (reporting results of survey distributed at the American Headache Society meeting in 2007, where 41% of respondents favored requiring more than three failed treatments in order for a patient to be consider refractory).

The evidence cited by Patent Owner to support that a POSA would have understood "refractory" to mean a "nonspecific resistan[ce] to treatment" does not support its position. For example, Exhibit 1053 (cited at PO Resp. 53), includes as "Box 2," an inset setting forth the "EHF [European Headache Federation] criteria for refractory chronic cluster headache." Ex. 1053, 3. Consistent with the definitions provided by Petitioner, it includes the requirement that the patients have "[f]ailed consecutive prophylactic treatment trials with at least three agents that showed efficacy over placebo in randomized controlled studies, used at the maximum tolerated dose over a sufficient period of time." *Id.* Plainly, this definition requires more than a "nonspecific resistan[ce] to treatment." The other reference cited by Patent Owner is to Weatherall (Ex. 1059), which Patent Owner points out is directed to failure of acute therapies. PO Resp. 46–47 ("In Weatherall, however, 'refractory to treatment' was defined by prior failure of 'standard *acute*' therapies, which are different from the classes of *preventative* treatments in the claimed method."). Patent Owner does not explain how or why this supports interpreting "refractory" in the Teva Press Release (or in Exhibit 1023), which discuss preventative treatments, to mean a "nonspecific resistan[ce] to treatment." *Id.* at 52.

In connection with its discussion of claim construction, Patent Owner asserts that "[a]bsent a specific definition in the art, . . . a POSA would have understood the word 'refractory' (unaccompanied by a proposed definition) to generally mean 'resistant to treatment,' without any implied conditions or degree or severity." PO Resp. 11. As support, Patent Owner cites two dictionary definitions for the term "refractory." *Id.* ¶ 51 (citing Ex. 2003 and 2006); *see also* Ex. 2037 ¶ 55 (Dr. Grosberg's testimony on the meaning of the term "refractory"). We find Petitioner's evidence as to how the POSA would have understood "refractory migraine" more persuasive because it cites articles specific to migraines while Patent Owner and Dr. Grosberg cite generic dictionary definitions of the term "refractory" that are not specific to any medical condition.

We give some weight to Patent Owner's arguments that the Teva Press Release is based on clinical trials that were not designed to study the claimed patients and do not provide data for such patients. PO Resp. 52. However, we cannot completely discount Patent Owner's characterization of these studies as showing that fremanezumab is effective in "highly refractory patients," particularly given that others appear to have reached the same conclusion (*See, e.g.*, Ex. 1023, 7 (concluding that anti-CGRP antibodies "are effective for relatively refractory headaches" based on data from the fremanezumab clinical trial reflected in Exhibits 1038 and 1058)).

In sum, we find that the clinical trials of fremanezumab, and the conclusion drawn from those trials in the Teva Press release, provide some support for the expectation that fremanezumab would be successful in treating the claimed patient groups.[29] This, in turn, helps to support an expectation that galcanezumab would be successful in treating the claimed patients groups. *See* Ex. 1101 ¶¶ 166, 170–174 (Dr. Evers's credible testimony that "[t]he results of the erenumab and fremanezumab studies supported the expectation that efficacy in treating 'RM' patients resulted from these agents effectively targeting CGRP generally, not specifically the CGRP receptor or ligand.").

[29] This would be true even if we were to interpret the term "highly refractory" in the Teva Press Release as to mean highly resistant to treatment, because, even construed as Patent Owner suggests, the Teva Press Release supports that fremanezumab specifically, and mAbs by extension, may provide relief to patients who are more difficult to treat than the general migraine population.

c. *Teachings in the art regarding the mechanisms of action of anti-CGRP antibodies and of other migraine treatments*

Petitioner contends that anti-CGRP antibodies have a mechanism of action that is distinct from that of the "commonly used preventative medications recited by the Patent's 'clusters' and 'selecting' limitation." Pet. 37–38. Because they have a different mechanism of action, according to Petitioner, "a POSA would have expected that galcanezumab would treat migraineurs independent of whether they used or failed conventional treatments that targeted non-CGRP pathways, e.g., topiramate (GABA inhibitor) or propranolol (beta blocker)." *Id.* at 38 (citing Ex. 1101 ¶¶ 177–178).

Patent Owner argues that the opposite is true: "a POSA would have understood that the refractory migraine patients of the claimed method—who already tried and failed *at least two different* classes of preventive medications—would have no expectation of success with a third treatment." PO Resp. 53. Patent Owner further argues that these patients had "already demonstrated a pattern of failure across different classes of medications with different mechanisms of action." *Id.* Finally, Patent Owner argues that "Petitioner points to nothing about anti-CGRP antagonist antibodies that would give rise to a greater expectation of success than any other third or subsequent treatment." *Id.* We disagree.

For the reasons discussed *supra* § II.F.2.a.iii, the evidence supports: 1) that anti-CGRP pathway mAbs are unlike traditional treatments in the sense that they block the CGRP pathway by specifically binding CGRP or the CGRP receptor, 2) that this difference in mechanism of action was recognized in the art, and 3) that anti-CGRP mAbs were treated as novel and different than prior treatments. Patent Owner's argument is unpersuasive for these reasons alone. In addition, we credit the well-supported testimony of Dr. Evers explaining why certain traditional treatments are different from anti-CGRP mAbs:

> Although the importance of CGRP in migraine pathophysiology was long recognized, none of the prophylactic migraine treatments available as of the earliest possible priority date of the '161 patent specifically targeted it. For instance, anticonvulsants (such as topiramate and valproic acid) were thought to act on migraine by regulating glutamate activity and gamma-aminobutyric acid (GABA). EX1019 (DeMaagd 2008 II), 3. Tricyclic antidepressants are "thought to involve the inhibition of central cortical depression and sympathetic activity 90 associated with migraine pathophysiology." EX1019 (DeMaagd 2008 II), 2. And, although not entirely understood, beta-blockers were believed to function via "modulation of the adrenergic nervous system and an influence on cranial blood vessels." EX1019 (DeMaagd 2008 II), 1.

Ex. 1101 ¶ 176.

Based on their different mechanism of action, Dr. Evers concludes that "a POSA would have reasonably expected that patients that had failed these other treatments could nonetheless be treated with compounds targeting CGRP directly." *Id.* ¶ 177. Dr. Evers's testimony is consistent with the multiple articles discussed *supra* § II.F.2.a.i suggesting that anti-CGRP mAbs have the potential to treat refractory patients. *See, e.g.*, Ex. 1037, 1; Ex. 1072, 74; Ex. 1041, 2; Ex. 1023, 7; *see also* Ex. 1053, 13. At least one of these reference attributes this potential their distinct mechanism of action. Ex. 1023, 7 (explaining that anti-CGRP mAbs "are effective for relatively refractory headaches" and that this "was not a serendipitous discovery" because there "were developed based on the pathophysiologic concept that the trigeminovascular system and CGRP have a key role in the development of migraine pain").

Dr. Evers also testifies, with citation to consistent supporting evidence that combination therapy with "then-available prophylactics had shown little to no success," but that fremanezumab "demonstrated an effect in patients allowed to remain on preventative treatments." Ex. 1101 ¶¶ 182–183 (citing Ex. 1038). This suggests that fremanezumab works by a different mechanism of action than the drugs with which it was combined. Ex. 2037 ¶ 158 (Dr. Grosberg's testimony regarding the "common knowledge in the prior art suggesting using medications having different mechanisms of action in combination therapy"); PO Resp. 40–41 ("A POSA would have known that standard combination therapy comprises agents from different classes and with different mechanisms of action."). According to Dr. Evers, this reinforced the expectation that targeting CGRP with a mAb would treat the claimed patient population. Ex. 1101 ¶ 181.

In sum, the evidence supports that galcanezumab's mechanism of action, which is distinct from that of commonly-used preventative medications, would have helped to foster a reasonable expectation of success.

d) Clinical trials that did not exclude refractory patients

Petitioner argues that plans for "numerous anti-CGRP mAbs clinical trials that allowed for inclusion of patients that failed two preventives further reinforced the reasonable expectation of success in treating the 'select[ed]' [refractory migraine] patients with galcanezumab.'" Pet. 39. As support Petitioner references the "eligibility criteria for completed galcanezumab, erenumab, and fremanezumab trials." *Id.* Petitioner also cites Dr. Evers testimony that "[a] POSA would have understood that initiating a Phase II clinical trial requires that the study sponsor provide convincing rationale for the study and anticipated efficacy, from which a POSA would understand there to have been a reasonable expectation of efficacy." Ex. 1101 ¶ 184.

Patent Owner argues that Petitioner's argument is "circular" and assumes its own premises. PO Resp. 55. According to Patent Owner, "[c]linical trials report failures in various patient population all the time – especially refractory patients." *Id.* at 56.

We have already discussed expectations fostered by each of the clinical trials referenced by Petitioner. *See supra* §§ II.F.3.b. We will not repeat that discussion here.

e) Conclusion regarding reasonable expectation of success

We find that Petitioner has established, by a preponderance of the evidence, that a POSA would have reasonably have expected galcanezumab to be successful when used in place of erenumab in Sun's method, which, as discussed *supra* § F.1, includes treating the claimed patients. This finding is strongly supported by Phase II clinical studies teaching that galcanezumab is effective in treating the general migraine population as well as a *post hoc* analysis supporting that galcanezumab was effective in treating patients with a history of failure to preventative treatments. *See supra* § II.F.3.a. Evidence supporting the efficacy of erenumab in patients who had failed prior treatments, particularly Tepper, also supports this finding. *See supra* §§ II.F.b.i–iii. So too does evidence supporting that fremanezumab is effective in the general migraine population and may be effective in "highly refractory" patients. *See supra* § II.F.b.iv. Finally, our conclusion that the POSA would reasonably have expected success using galcanezumab in

Sun's methods finds support in mAb's distinct mechanism of action. *See supra* § II.F.c.

4. Does using galcanezuab in Sun's methods meets the limitations of claim 1 for all three patient groups

Having determined that Sun discloses or suggests treating the claimed patient groups (*see supra* § II.F.1), and supports using galcanezumab in Sun's methods with a reasonable expectation of success (*see supra* §§ II.F.2–3), we now consider whether using galcanezumab in Sun's methods meets each of the limitations of claim 1.

d) Preamble

The preamble of claim 1 recites "[a] method of treating or preventing migraine in a subject having refractory migraine." For the reasons discussed *supra* § II.F.1, Sun discloses or suggests treating each of Group I, Group II, and Group III patients respectively under our construction of the term "refractory. Using galcanezumab to treat these patients, as proposed by Petitioner, meets the language of the preamble. According, we find that Petitioner has established, by a preponderance of the evidence, that the cited art teaches or suggests the language recited in the preamble.

e) The "selecting" limitation

Claim 1 next recites "selecting a subject who has been treated with two or more different preventative migraine treatments wherein at least one of the preventative migraine treatments is selected from topiramate, carbamazepine, divalproex sodium, sodium valproate, valproic acid, divalproex, flunarizine, candesartan, pizotifen, amitriptyline, venlafaxine, nortriptyline, duloxetine, atenolol, nadolol, metoprolol, propranolol, bisopropol, timolol, onabotulmumtoxin A, lisinopril, and oxeterone."

As discussed *supra* § II.F.1.a–c, Sun discloses treating each of Group I, Group II, and Group III patients respectively under our construction of the term "refractory. Sun provides a list of prior treatments and classes of treatments that its patients may have failed or been intolerant to. Ex. 1006 ¶¶ 67–68. Many of the treatments on Sun's list overlap with those called out in our claim construction. *Supra* § II.C.1 (our claim construction); *compare* Ex. 1006 ¶¶ 67–68, *with* Ex. 1002, 19:25–60.

For Group I and Group II patients, Petitioner contends, and Patent Owner does not dispute, that failed treatment – whether due to a lack of clinically meaningful response or intolerable side effects – "presuppose[s] having been treated with the specified medications." Pet. 27. We agree. Thus, Sun's disclosure or suggestion to treat Group I and Group II patients, respectively, involves "selecting" patients that were previously treated with treatments identified in our claim construction. Ex. 1101 ¶¶ 125–126.

Sun also discloses treating Group III patients, and the treatments that such patients are contraindicated for are the same as those Petitioner relies upon for the Group I and Group II patients. Ex. 1006 ¶¶ 67–68. However, as Patent Owner points out, and as Petitioner does not dispute, a patient that is contraindicated for a particular drug would not be treated with that drug. PO Resp. 14. Thus, unlike treating Group I and Group II patients, treating a contraindicated patient, does not presuppose "selecting" patients that have previously been treated with the specified medications.

Petitioner addresses this problem by arguing first that "*Sun* teaches and provides motivation for treating [Group III] patients that were contraindicated from two different classes of medications," and second that "*Sun* teaches and provides motivation for 'selecting' [Group III] patients for treatment with anti-CGRP mAbs, including galcanezumab, that have previously used conventional preventative treatments, such as antiepileptics, beta-blockers, and anti-depressants as claimed." Pet. 41. Thus, we understand Petitioner to argue that Sun suggests treating patients that are contraindicated to two drugs and that have been administered two different non-contraindicated drugs. *See* Pet. Reply, 25 ("Teva . . . ignores that there is no conflict in 'selecting' a patient that has been treated with two drugs or classes (e.g., anticonvulsant and an angiotensin II receptor antagonist) who is also contraindicated for two other drugs or classes (e.g., beta-blockers and tricyclic antidepressants).").

As to the motivation for "selecting" patients who have been treated with non-contraindicated drugs, Petitioner contends that the POSA would have selected patients using conventional preventative treatments, such as those identified in our claim construction, because Sun teaches that using anti-CGRP mAbs avoids side effects associated with other migraine prophylactic treatments. Pet. 41–42 (citing Ex. 1006 ¶¶ 14, 250; Ex. 1101 ¶ 190). Petitioner cites the testimony of Dr. Evers, who explains that, based on this advantage, "a POSA would have understood *Sun* to teach and suggest the use of anti-CGRP mAbs as a replacement medication for patients previously treated with conventional preventive treatments, such as antiepileptics, beta-blockers, and anti-depressants." Ex. 1101 ¶ 190.

In its Response, Patent Owner does not challenge any aspects of Petitioner's positions on Group III patients, other than arguing that the "selecting" clause excludes contraindicated patients from the scope of the claim. *See generally* PO Resp. This argument is not persuasive for the reasons discussed *supra* § II.C.2.

In its Sur-reply, Patent Owner asserts that it "showed that Petitioner's cited art does not teach or suggest treating these [contraindicated] patients or provide any motivation to treat these patients with a reasonable expectation of success." PO Sur-reply 20 (citing Ex. 2037 ¶¶ 170–183, 200–203, 237–238, 249–250, 252–276). As discussed, *supra* § II.F.1.c, this argument is waived and violates our rules prohibiting incorporation by reference. Accordingly, we do not consider it.

Even if we were to consider this argument, we would not find it persuasive because Patent Owner's waived argument relates not to the issue at hand – whether Sun suggests treating patients that are contraindicated to two drugs and that have been treated with two different non-contraindicated drugs – but to whether Sun discloses treating contraindicated patients. *See supra* § II.F.1.c.

Considering the evidence with respect to the "selecting" limitation as applied to Group III patients, we find that the preponderance of the evidence supports Petitioner's position. Sun teaches treating contraindicated patients with enernumab. Ex. 1006 ¶ 66. Sun also teaches that its anti-CGRP mAb, "does not substantially cause an adverse side effect associated with other migraine prophylactic treatments, including adverse side effects associated with antiepileptics, beta-blockers, and anti-depressants." *Id.* ¶ 14. Absent persuasive evidence to the contrary, we agree with Petitioner and Dr. Evers that this teaching provides motivation to "use . . . anti-CGRP mAbs as a replacement medication for patients previously treated with conventional preventive treatments, such as antiepileptics, beta-blockers, and anti-depressants," including in patients that are contraindicated to two medications that they are not currently taking. Ex. 1101 ¶ 190. Treating such patients would involve "selecting a patient who has been treated two or more different preventative migraine treatments" selected from the group recited in claim 1.

In sum, we find that Petitioner has established, by a preponderance of the evidence, that Sun discloses the "selecting" limitation as applied to each of the claimed Group I, Group II, and Group III patients, respectively.

f) The "administering" limitation

The final limitation of claim 1 recites "administering to the subject a therapeutically effective amount of a humanized monoclonal anti-calcitonin gene-related peptide (CGRP) antagonist antibody comprising the amino acid sequence of the heavy chain variable region set forth in SEQ ID NO: 63 and the amino acid sequence of the light chain variable region set forth in SEQ ID NO: 62."

As an initial matter, there does not appear to be any dispute, and thus we find, that the heavy chain variable region set forth in SEQ ID NO: 63 and the amino acid sequence of the light chain variable region set forth in SEQ ID NO: 62 correspond to galcanezumab. Ex. 1101 ¶ 133. For the reasons discussed *supra* § II.F.2, the POSA would have been motivated to use galcanezumab in Sun's method of treating the claimed Group I, Group II, and Group III patients, respectively. For the reasons discussed *supra* §§ II.F.3, the POSA would reasonably have expected success in doing so. According, we find that Petitioner has established, by a preponderance of the evidence, that the cited art teaches or suggests the language recited in the "administering" limitation.

G. Ground 1, claims 2–30

Claims 2–30 depend from and further limit claim 1 by reciting additional requirements: that the subject have taken a specific number and/or type of prior treatments (claims 2–7), that the subject have been treated with two or more treatments within the 10 years prior to being administered the mAb (claim 8), that the subject have been diagnosed with episodic (claim 9) or chronic migraine (claim 10), that the subject experience migraine with (claim 11) or without aura (claim 12), that the patient be administered a particular dose of mAb (claims 13, 14, 15, 16, and 17), that the patient be administered a particular dose by two separate injections (claim 18), that the mAb be administered once per month (claim19), that the mAb be administered as a liquid at a particular concentration (claims 20 and 21), that a particular volume of mAb be administered (claims 22–24), that the mAb be administered by a particular device (claim 25), that the mAb achieve a specific reduction in headache days (claim 26), that the mAb be administered in combination with an acute headache medication (claim 27) and that such combination decrease the monthly amount of the acute headache medication administered (claim 28), that the mAb is administered subcutaneously (claim 29), and that the mAb comprise specific heavy and light chain amino acid sequences (claim 30).

Petitioner contends that the limitations recited in claims 2–30 are disclosed in Sun, Dodick, and/or Sharma.[30] Pet. 42–45, 55–66. Petitioner

[30] Patent Owner argues that Petitioner failed to challenge dependent claims 6, 11–12, 14–15, 19–23, and 30 in Ground 1. PO Resp., 56–57. Patent Owner is mistaken. The Petition includes the following unambiguous heading: "Ground 1: Claims 1-30 Would Have Been Obvious Over *Sun*, *Dodick*, and *Sharma*." Pet. 18. Under this heading, as part of the discussion of Ground 1, the Petition states: "[c]laims 2-30 are obvious for the same reasons addressed in Ground 2 based on *Dodick* and *Sharma* alone." *Id.* at 42. Thus, in an apparent effort to avoid repeating the same analysis twice, and to economize on space, Petitioner addressed the application of Sun's teachings to certain dependent claim as obvious at pages 42–45 (which under the heading "Ground 1") and the application of Dodick and Sharma's teachings to certain dependent claims at pages 55–66 (which under the

provides documentary and testimonial evidence to back its assertions. *Id.*; *see also* Ex. 1101 ¶¶ 193–223, 262–352.

Patent Owner does not challenge Petitioner's showing with respect to dependent claims 2–30 other than to assert, as to all twenty-nine claims, that "Petitioner focuses solely on claim elements purportedly disclosed in the art, without *any* discussion of why a POSA would have allegedly modified the references or had a reasonable expectation of success in doing so." PO Resp. 56. This argument lacks merit. For example, for claims 2 and 3, which require that the subject have been treated with "two to four" (claim 2) different preventative migraine treatments, and for claims3–7, which recite *Markush* groups of failed treatments, Petitioner asserts:

> [Claims 2–7] read on the average and commonly treated migraine patient, who had been previously treated with three or four treatments. EX1015, 1-2; EX1042, 8.
>
> In addition to reasons discussed above, the motivation and expectation of success for treating the "selected" patients of claims 2-7 was reinforced by the art as [a] whole. For example, using anti-CGRP mAbs to treat patients having previously taken two different migraine prophylactics, including patients that have failed two or more "migraine prophylactic agents… include[ing]… propranolol… metoprolol… divalproex, sodium valproate… topiramate… amitriptyline… and botulinum toxin type A." EX1006, [0067]. The motivation and expectation were reinforced by a POSA's understanding that, regardless of the

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heading "Ground 2"). As Petitioner made clear that it was challenging claims 1–30 as part of Ground 1 (Pet. 19) and further made clear that Ground 1 also relied on the reasoning discussed in connection with Ground 2 (Pet. 42), we find no merit in Patent Owner's argument that Petitioner's challenge is not sufficiently specific under 35 U.S.C. § 312(a)(3). Put another way, because the Petition was clear as to what assertions apply to what grounds, we do not fault Petitioner for not separately repeating verbatim assertions that apply to multiple grounds each time they were relevant to a particular ground.

> number of drugs previously taken, efficacy would have been expected because migraine patients share the same physiopathology. EX1101, ¶274; see In re Montgomery, 677 F.3d at 1382-83. Pet. 55–56. Contrary, to Patent Owner's assertion, Petitioner does not focus solely on where claim elements are recited in the prior art; Petitioner clearly explains both why the POSA would have been motivated to treat the patients recited in claims 2–5 and why the POSA would reasonably have expected success in doing so.

Similarly, for claim 8, which requires that the subject have been "treated with two or more different classes of preventative migraine treatments within the 10 years prior to administering the monoclonal antibody," Petitioner asserts:

> Dodick rendered this claim obvious, disclosing treatment of patients that had used other preventive treatments as recently as 31 days (or 121 days for botulinum toxin), well within the ten-year limitation. EX1003, 2; EX1101, ¶¶275-278. Indeed, botulinumtoxin was approved for migraine prevention less than ten years before Dodick's publication, such that its contemplated prior use fell within claim 8. EX1070, 2 (n.13-14); see also EX1036, 2 (anti-CGRP mAb study allowing patients that discontinued other preventives); EX1038, 2 (same); EX1039, 3 (same).
>
> The motivation and expectation of success were reinforced by the art, which showed that migraine patients were prescribed prophylactics until achieving clinical success. EX1101, ¶¶277-279; EX1020, 4-5; EX1005, 7 ("Often trial and error [was] needed"). As standard practice suggested not leaving patients untreated, a POSA would have been motivated to treat patients with galcanezumab soon after they had failed other preventives. EX1101, ¶278

Pet. 56–57. As with claims 2–7, discussed above, Petitioner does not rest solely on the identification of where claim elements can be found in the prior art. Petitioner provides articulated reasoning with rational underpinning to support it proposed finding of obviousness. The same holds true for the remainder of the challenged dependent claims. Accordingly, we are not persuaded by Patent Owner's argument that the Petition lacks "*any* discussion of why a POSA would have allegedly modified the references or had a reasonable expectation of success in doing so." PO Resp. 57.

We have reviewed Petitioner's contentions and determine that Petitioner has shown, by a preponderance of the evidence, that claims 2–30 would have been obvious over the combination of Sun, Dodick, and Sharma. *See In re NuVasive, Inc.*, 841 F.3d 966, 974 (Fed. Cir. 2016) ("The Board, having found the only disputed limitations together in one reference, was not required to address undisputed matters."); *see also* Paper 12 at 8 (emphasizing that "any arguments not raised in the response may be deemed waived").

H. Ground 2

Petitioner argues claims 1–30 of the '161 patent would have been obvious over Dodick and Sharma. We have already determined claims 1–30 are unpatentable based on Petitioner's Ground 1. *See supra* § II.F. and II.G. Therefore, we need not, and to conserve the Board's resources we do not, reach Ground 2. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner "is entitled to a final written decision addressing all of the claims it has challenged"); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App'x 984, 990 (Fed. Cir. 2020) (non-precedential) (recognizing that the "Board need not address issues that are not necessary to the resolution of the proceeding" and, thus, agreeing that the Board has "discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims").

III. CONCLUSION

Petitioner has established by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Sun, Dodick, and Sharma.

IV. PATENT OWNER'S MOTION TO EXCLUDE

Patent Owner filed a Motion to Exclude seeking to exclude Exhibits 1040, 1041, 1065, 1073, 1075, 1088, 1089, 1208, 1209, 1210, 1220, and 1221 as inadmissible hearsay under FRE 801 and 802. Paper 36, 2 ("Motion" or "Mot."). Patent Owner also sought to exclude the portions of Dr. Evers's declaration that rely on the allegedly inadmissible exhibits. *Id.* Petitioner filed an Opposition (Paper 37, "Opposition" or "Opp.") and Patent Owner filed a Reply (Paper 38, "Mot. Reply"). We deny Patent Owner's motion in part and dismiss Patent Owner's motion in part as moot.

A. Exhibits 1065, 1073, 1088, 1089, 1220, and 1221

We do not rely on Exhibits 1065, 1073, 1088, 1089, 1220, and 1221 in our Final Written Decision. Accordingly, we dismiss Patent Owner's Motion as it relates to these exhibits.

B. Exhibit 1040

Exhibit 1040 ("Goadsby") is prior art describing the results of Sun's proposed Phase III trial. Ex. 1040, 15–16. In relevant part, it states:

> Robust treatment effects were observed for both 70 mg and 140 mg erenumab in subjects who had previously failed preventative migraine treatments. For 140 mg, effects were numerically greater in this sub-population than in the overall trial population, and as in the overall population, erenumab 140 mg showed numerically greater efficacy than erenumab 70 mg. These results suggest that erenumab may have particular utility in this subgroup of patients.

Id.

Patent Owner argues that Petitioner relies on Goadsby to show that erenumab had "shown success in patients with a history of treatment [] failure" as well as to show that it achieved "robust treatment effects . . . in subjects who had previously failed preventive migraine treatments' and suggested 'particular utility in this subgroup of patients.'" Mot. 3. We disagree. Petitioner relies on this Goadsby, not for the truth of its assertions. Rather, Petitioner relies on Goadsby for the non-hearsay purpose of showing its effect on the hypothetical person of ordinary skill in the art. *See* Opp. 5 (describing Petitioner's use of Goadsby). We rely on it in the same manner. *See supra* § II.F.3.b.iii (discussing whether Goadsby supports that the POSA would have expected success in using galcanezumab in Sun's methods). Accordingly, we deny Patent Owner's Motion as it relates to Exhibit 1040.

C. Exhibit 1041

Exhibit 1041 ("the Teva Press Release") is prior art characterizing the results of two clinical studies of fremanezumab. In relevant part, it states:

> "The collective data generated from these studies herald promise for millions of people who suffer from episodic and chronic migraines, a disease with substantial implications and unmet needs," stated Marcelo E. Bigal, Teva's Head of Global Clinical Development for Migraine and Headaches [and a named inventor on the '161 patent]. "The very fast onset of preventive response, seen after a single dose of therapy, along with the impressive decrease in migraine days, amongst such highly refractory patients, may bring us a step closer to provide widespread relief to people who suffer from chronic and episodic migraine."

Ex. 1041, 2 (emphasis added).

Patent Owner argues that Petitioner improperly relies on Exhibit 1041 to prove that "fremanezumab … show[ed] success in patients with a history of treatment failure" (id. at 5-6) and to prove that "fremanezumab, which—unlike traditional preventives—showed efficacy as a combination therapy." Mot. 3. We disagree. As with Goadsby, Petitioner relies on the Teva Press Release not for the truth of the matter asserted in the reference at issue, but for the non-hearsay purpose of showing the effect of the reference on the hypothetical person of ordinary skill in the art. *See* Opp. 6 (describing Petitioner's use of the Teva Press Release). We rely on it in the same manner. *See supra* § II.F.3.b.iv (discussing whether the Teva Press Release supports that the POSA would have expected success in using galcanezumab in Sun's methods). Accordingly, we deny Patent Owner's Motion as it relates to Exhibit 1041.[31]

D. Exhibits 1208, 1209, and 1210

Exhibits 1208 and 1209 are FDA guidance documents and Exhibit 1210 is guidance from the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). All three of these Exhibits provide definitions of the term "adverse event." Patent Owner argues that "Petitioner . . . uses these references to prove that the definitions asserted therein are the true definitions that should be used to interpret the claims." Mot. 6. We disagree. As with Goadsby and the Teva Press Release, Petitioner relies on Exhibits 1208–1210 not for the truth of the matter asserted, but to show how the POSA would have understood the term "adverse event." *See* Opp. 10 (describing Petitioner's use of Exhibits 1208, 1209, 1210). We rely on them in the same manner. *See supra*

[31] The Teva Press Release is also a statement authorized and made by Patent Owner and, therefore, not hearsay under Fed. R. Evid. 801(d)(2).

§ II.F.1.b. Accordingly, we deny Patent Owner's Motion as it relates to Exhibits 1208–1210

E. Dr. Evers's Declaration

Patent Owner argues that we should exclude ¶¶ 25 n.3, 225, and 235 of Dr. Evers's Declaration because they "rely on Exhibits 1004, 1087, and 1088 for improper hearsay purposes." We deny Patent Owners Motion as it relates to Dr. Evers's declaration for three reasons. First, as to ¶¶ 25 n.3 and 225, Patent Owner seeks to exclude the challenged testimony on the basis that it relies on Ex. 1004 and Ex. 1087, which Patent Owner asserts are hearsay. But Patent Owner did not move to exclude either exhibit. Second, for the reasons discussed in Petitioner's Opposition (Opp. 12–13) we agree that the challenged testimony relies upon Exhibits 1004, 1087, and 1088 for non-hearsay purposes. Third, even if the cited exhibits were hearsay, FRE 703 permits an expert to rely on the kinds of facts and data an expert in a field would reasonably rely on, regardless of whether those facts or data are themselves admissible. FRE 703 ("If experts in the particular field would reasonably rely on those kinds of facts or data in forming an opinion on the subject, they need not be admissible for the opinion to be admitted"); *i4i Ltd. P'ship v. Microsoft Corp.*, 598 F.3d 831, 852 (Fed. Cir. 2010).

F. Conclusion

We deny Patent Owner's motion as to Exhibits 1040, 1041, 2008–2010, and Dr. Evers's testimony. We dismiss as moot, Patent Owner's motion as it relates to Exhibits 1065, 1073, 1088, 1089, 1220, and 1221.

V. CONCLUSION

Based on the evidence presented with the Petition, the evidence introduced during the trial, and the parties' respective arguments, Petitioner has established by a preponderance of the evidence that claims 1–30 are unpatentable as obvious over the combination of Sun, Dodick, and Sharma.[32] We decline to reach the merits of Petitioner's Ground 2, which asserts that claims 1–30 of the '161 patent would have been obvious over Dodick and Sharma, because we have already determined that claims 1–30 are unpatentable based on Petitioner's Ground 1. We deny Patent Owner's Motion to Exclude as it relates to Exhibits 1040, 1041, 2008–2010, and Dr. Evers's testimony. We dismiss as moot, Patent Owner's Motion to Exclude as it relates to Exhibits 1065, 1073, 1088, 1089, 1220, and 1221.

In summary:

Claims	**35 U.S.C. §**	**Reference(s)/ Basis**	**Claims Shown Unpatentable**	**Claims Not Shown Unpatentable**
1–30	103(a)	Sun, Dodick, Sharma	1–30	
1–30	103(a)	Dodick, Sharma[33]		
Overall Outcome			1–30	

[32] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding after the issuance of this Final Written Decision, we draw Patent Owner's attention to the April 2019 Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. §§ 42.8(a)(3), (b)(2).

[33] We do not reach this ground for the reasons discussed *supra* § II.H.

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–30 in the ’161 patent are determined to be unpatentable;

FURTHER ORDERED that Patent Owner’s Motion to Exclude is denied as it relates to Exhibits 1040, 1041, 2008–2010, and paragraphs 25 n.3, 225, and 235 of Dr. Evers’s Declaration (Ex. 1092);

FURTHER ORDERED that Patent Owner’s Motion to Exclude is dismissed as moot as it relates to Exhibits 1065, 1073, 1088, 1089, 1220, and 1221; and

FURTHER ORDERED that because this is a final written decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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